Twenty-five conjectures from a local–global random model, with machine verification and adversarial audit

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Abstract

We state twenty-five conjectures in elementary and analytic number theory, each derived from the local–global random model of the primes: Cramér's model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events. Each statement is calibrated to the strongest form the heuristic supports and no stronger. Every computable constant is computed from its definition, with local admissibility asserted programmatically; every statement was tested against exact counts, with success measured by the *shape* of agreement (predicted constant, square-root residuals, no drift) rather than by the size of the range searched. The statements were then subjected to a three-layer adversarial audit: an independent literature search for prior art, scripted refutation attempts a decade beyond the original bounds, and clean-context re-derivations of constants and counts. The audit refuted one conjecture as first stated—the exceptional set for $n = p + k^3$ contains the infinite family of cubes k^3 with $3k^2 - 3k + 1$ composite, an algebraic obstruction the probabilistic accounting missed—and it appears here restated, with the obstruction promoted to a theorem and the original error documented; a later round of review caught a priority failure (a statement we carried as new had been published), which is likewise documented rather than erased. We are deliberately conservative about novelty, and structure the paper accordingly: after successive review rounds every attributed benchmark was either lifted to a family or mechanism law or replaced outright, so that each of the twenty-five slots now claims content of its own—while every retired classical statement is retained *inside* its successor slot as an attributed calibration remark, since the benchmarks are what make the novel layer's constants credible. The claimed content includes: a prime-power contamination calculus for pattern races—twin races modulo 5 and 8 driven by the pairs $(q^2 - 2, q^2)$, fresh cousin, sexy-pair, and triplet predictions derived from the calculus before testing, a Goldbach lane race with a verified internal null lane, a Stern-representation lane race whose null classes are provable by direct congruence, and an algebraic null control; the measured sub-diffusivity of balanced races (universal negative step correlations tied to the Lemke Oliver–Soundararajan repulsion); canonical ordering deficits for least primes in progressions and for least Goldbach summands; derived covariance kernels for moving-window residual fields, with the diagonal deficit reduced to a pinned singular-series average whose growth already exceeds the single-prime Montgomery–Soundararajan term; distribution laws for the constants of the quadratic-pair and cubic-shift families, with exact mean-value lemmas; an entanglement-aware local-global law for $n^2 + 2^n$ with an exact-rational CRT factorization; a boundary trichotomy for polynomial ladders; a multibase subtorus law for Fermat quotients; and singular-series waiting-time refinements for first prime gaps. Where an audit refuted

*All constants, counts, and audit results in this paper are reproducible from the accompanying code repository: `run_all.py` regenerates every number. Primality of integers beyond 3.3×10^{24} was tested with Baillie–PSW; all smaller primality claims are deterministic (fixed-base Miller–Rabin).

our own drafts—a false exception census, a wrong occupation clock, a naive joint constant contradicted by a catalogued Fibonacci–Lucas twin at index 148091—the refutations are documented in place.

Contents

1 Introduction

A good conjecture in the theory of numbers is not merely a statement that has resisted counterexample. It is the *prediction of a model*: the assertion that the primes, beyond their local structure, behave like a random set of density $1/\log n$. The discipline this imposes is exacting. A conjecture worth stating should (i) survive every congruence and size obstruction, which is a computation, not a hope; (ii) come with a quantitative asymptotic whose constant is derived, not fitted; (iii) demand exactly the fluctuations the model earns—square-root noise with logarithmic corrections—and no fewer (the Mertens conjecture died of demanding fewer); (iv) sit inside the standard hierarchy of conjectures (Hardy–Littlewood k-tuples $\subset$ Schinzel's Hypothesis H $\subset$ Bateman–Horn), so that its failure would propagate; and (v) be falsifiable instance-by-instance by computation. Numerical range is the least important column in the ledger: most deep phenomena drift at the rate $\log\log x$, and $\log\log 10^{18} \approx 3.7$, so verification "to 10^{18}" is weak evidence by itself. What persuades is the *shape* of the agreement—a count that tracks a derived constant through several decades with residuals that look like noise.

The twenty-five statements of this paper were produced systematically to those specifications, together with the audit that the specifications demand. They are organized in four blocks: Bateman–Horn systems (§3), barely-divergent sparse sequences (§4), representation problems with Borel–Cantelli accounting (§5), and statistical laws of the prime sequence (§6). Section 7 describes the adversarial audit: what was attacked, what survived, and in one case what did not. We consider the audit trail part of the mathematical content. Its headline is instructive: the one outright refutation (Conjecture 13, first version) was produced not by bad luck but by a density-zero algebraic family—the cubes—invisible to a probability model that integrates over density. This is precisely the classical lesson of admissibility, relearned here in public.

Two further audit outcomes deserve emphasis. First, an independent literature layer identified every statement that already existed in essence—some verbatim (Dubner's twin-sum conjecture, the Stern list for $p+2k^2$, Caldwell–Gallot's $e^{\gamma}\log N$ laws), one with an exact constant catalogued as an OEIS entry (A188596), and in two instances statements we had carried as new were found already published (Finding 2, twice). After successive replacement rounds no such statement occupies a slot on its own any longer: each is retained *inside* its successor slot as an attributed calibration remark, restated in our uniform framework and re-verified at our bounds, with the attribution ledger in §8; every current slot's claimed content was novelty-searched at neighbourhood depth—defining objects, derived sequences, OEIS comments, and the abstracts and bodies of near-neighbour papers—the standard the Finding 2 failures taught us to demand. One replacement candidate was *refuted by this search before publication*: a Fibonacci–Lucas twin completeness list that OEIS A080327's catalogued index 148091 destroys, an episode now built into Conjecture 9 as its calibration clause. Second, several of our *first drafts* were corrected by data before any external audit: the naive Poisson claim behind Conjecture 20 fails at every accessible scale (the corrected second-order law is the conjecture); the first-guess bias direction in the twin race of Conjecture 21 was reversed by a mod-3 computation; the naive Exp(1) mean in Conjecture 22 is off by a factor that took three controls to localize. We report these corrections because the difference between a fitted claim and a derived claim is exactly what an audit is for.

Taxonomy: claimed content and its calibration layer

An external review of an earlier draft made a criticism we accept in full: an unrecorded specialization of Bateman–Horn is analogous to evaluating a classical special function at a new argument—worthwhile as data, but not a new conjecture in the conceptual sense. Six review rounds later, that criticism has been discharged by construction: no slot's claimed content is a bare specialization any longer. The taxonomy is now two-layered *within* each slot. The *calibration layer*—every previously-known statement this paper began from (Dubner's conjectures, the Stern list, the Caldwell–Gallot laws, Martin's refined Goldbach, Kourbatov's record framework, the classical single instances of each family)—survives as attributed remarks inside the slots, each with its derived constant re-verified at our bounds; the criteria document counts this hierarchy membership as a virtue, and the calibration layer is what makes the novel layer's constants credible. The *claimed layer* is graded honestly: mechanism-level content (the contamination calculus 21(v) with its instances 3, 7, 8, 14, 21, 25; the ordering deficits 15, 22; race sub-diffusivity 18; the entanglement law 11); derived second-order and distributional laws (1, 2, 12, 16(ii), 19(iii), 20, 23(iv), 24(ii)); and structural classifications (4, 5, 6, 9(i), 10(i), 13)—with framework attributions stated where an instance lives inside someone else's theory (Kowalski's for 2(ii), Montgomery–Soundararajan's for 12 and 20). Every statement admitted in the replacement rounds was novelty-searched at neighbourhood depth and verified at production scale before entering the paper. Within the research group we regard the prime-power contamination calculus (Conjecture 21(v)) and its mechanism family—21 and 25, the fresh cousin-race predictions it generated, and the negative control 8—together with Conjecture 22(ii) and the derived covariance kernels of 16(ii) and 24(ii) as the paper's core, joined by 20 and 4; Conjectures 9, 11 and 19 carry open modeling questions flagged in place.

Verification standards

All counts below are exact (sieves; deterministic Miller–Rabin for $n < 3.3 \times 10^{24}$; Baillie–PSW beyond, flagged as such). Every comparison reports the Poisson-normalized residual $z = (\mathrm{obs} - \mathrm{pred})/\sqrt{\mathrm{pred}}$. For sparse counts the cumulative comparison is contaminated at small x by the phantom mass of the main-term integral near its lower limit; in those cases we also report per-decade increments, which are clean. Constants are computed by truncated Euler products with the truncation wobble (difference between cutoffs) reported as an error bar; for conditionally convergent quadratic and cubic products this wobble, not the last digit, is the honest precision.

2 Notation

p, q denote primes; γ is Euler's constant, $\mathrm{e}^{\gamma} = 1.781072\ldots$; φ is Euler's totient; $\phi = (1+\sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \ldots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \ldots, f_k) \;=\; \prod_p \frac{1 - \omega(p)/p}{(1-1/p)^k}, \qquad \omega(p) = \#\{n \bmod p : \; p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) \;=\; \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \le N :$ all $f_i(n)$ prime$\} \sim C \cdot I(N)$ whenever $C \neq 0$; every conjecture in §3 is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product, not a substitute for that convergence statement. We write $\mathrm{Li}_k(x) = \int_2^x (\log t)^{-k}\, dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p \mid d,\, p>2} \frac{p-1}{p-2}$ for even d, with $2C_2 = 1.3203236\ldots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

For locally integrable F we write $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t)\, \frac{dt}{t}$ for the logarithmic mean; all race conjectures below state their drift and null clauses through $\mathcal{M}_x$, so that "no persistent leader" and "systematic surplus" are claims about a defined functional rather than about pointwise behaviour.

Admissibility ($\omega(p) < p$ for all p) is asserted programmatically during every constant computation. The check does real work: the natural-looking pair $\{n^2+n+1,\; n^2+n+3\}$ has $\omega(3) = 3$ and is rejected; the $k = 4$ branch of Conjecture 5 and the naive iterated chain behind Conjecture 6 were both rejected by the same machinery, and both rejections became part of the statements.

Novelty labels, from the independent literature audit (§8): **(a)** previously stated in essence; **(b)** classical family, quantified instance apparently unstated.

3 Bateman–Horn instances

The eight conjectures of this block are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it any longer:

1 and 2 are family laws with derived constant distributions, 4 and 5 carry structural classifications, 6 is a searched chain, and 3, 7, 8 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism lives in §6. Each slot retains its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1-\omega(p)/p)/(1-1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

Conjecture 1 (Uniform quadratic de Polignac; family law apparently unstated, the $d = 2$ instance is (a), cf. OEIS A080149, [32])**.** *For even $d \geq 2$ let $\pi_d^{\mathrm{q}}(N) = \#\{n \leq N :\ n^2+1,\ n^2+1+d$ both prime$\}$ and $C(d) = C(x^2+1,\ x^2+1+d)$. Then:* (i) *[uniform family law] for every fixed $B > 0$, $\pi_d^{\mathrm{q}}(N) = C(d)\, I_d(N)(1+o(1))$ uniformly over even $d \leq (\log N)^B$;* (ii) *[registered rate, strictly stronger] there is $\eta_B > 0$ with*

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2|d}} \Big| \frac{\pi_d^{\mathrm{q}}(N)}{C(d)\, I_d(N)} - 1 \Big| \ll_B (\log N)^{-\eta_B}$$

—an error clause that no version of Bateman–Horn supplies and which we register separately from (i), since its analytic content is independent (the sixth external review is right that an unspecified exponent must not be presented as part of the base conjecture); (iii) *[family statistic: the law of the constants] each local factor of $C(d)$ depends only on $d \bmod p$, and the factors at any* finite *set of primes are exactly independent as d varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via L^2-boundedness of the normalized partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of $C(d)$ over even $d \leq D$ converge, as $D \to \infty$, to* derived *Euler products; in particular*

$$\frac{1}{\#} \sum_{2|d \leq D} C(d) \to \bar{C} = 2.7447\ldots, \qquad \mathrm{sd}(C(d)) \to 1.6835\ldots,$$

and the empirical distribution of $C(d)$ converges to the law of the random product $\prod_p f_p(U_p)$ with independent uniform residues U_p. In particular ($d = 2$, the classical instance that formerly stood alone in this slot): $\#\{n \leq N :\ n^2+1,\ n^2+3$ both prime$\} \sim C(2)\, I(N)$, $C(2) = 2.954014\ldots$

This is the quadratic analogue of the uniform de Polignac law of Conjecture 16: not one count but the whole profile of constants must be matched at once, and the family, unlike its linear parent, has no known prior statement. Every even shift is admissible (for $p \geq 5$, $\omega(p) \leq 4 < p$; the classes mod 3 and 5 vary with d, which is what makes the $C(d)$-profile nonconstant, spanning a factor ≈3 over $d \leq 300$). At $d = 2$ the local analysis reads: n even, $n \not\equiv 0 \pmod 3$, $\omega(p) = 2 + (\frac{-1}{p}) + (\frac{-3}{p})$ for $p \geq 5$, and the conjecture implies infinitely many twin pairs $(m+1, m+3)$ with m a perfect square. Profile verification at $N = 10^6$ over all 150 even shifts $d \leq 300$: correlation 0.99976 between observed and predicted counts, regression slope 1.0022, residual mean $z = +0.24$ with spread 0.84 and $\max_d |z| = 2.28$—the shape-of-agreement standard applied across the family, not cherry-picked at one shift. Clause (iii) was verified by computing both sides independently: the derived Euler-product moments give mean 2.7447 and standard deviation 1.6835, against 2.7434 and 1.6726 measured over the 150 constants with $d \leq 300$ (ratios 0.9995 and 0.9935)—a family-level statistic whose constants are derived, not fitted, added at the sixth external review's suggestion. One question the family raises is recorded as open rather than conjectured: the maximal range $D(N)$ for which the uniform law can hold (the analogue of the Elliott–Halberstam question for this family—whether there is a sharp transition at some power scale N^δ rather than a logarithmic one).

Conjecture 2 (The cubic shift family: the distribution of its constants; instance-level law apparently unstated)**.** *For non-cube $a \geq 1$ let $C(a) = C(x^3 + a)$. Only $p \equiv 1 \pmod 3$ moves the product, and the local root count there takes* three *values as a varies mod p: $\omega = 3$ with probability $(p-1)/3p$ ($-a$ a nonzero cube), $\omega = 1$ with probability $1/p$ ($p \mid a$—the state an earlier draft's prose omitted, caught by the seventh review; the computations always carried it), and $\omega = 0$ with probability $2(p-1)/3p$; for $p \equiv 2 \pmod 3$ the cube map is a bijection and $\omega = 1$ identically. Then:* (i) *[mean-value one, elementary lemma] $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$ exactly at every p, so the derived mean of $C(a)$ is exactly 1—the cubic analogue of the Korevaar–te Riele mean-value-one theorem for linear prime-pair constants (arXiv:0806.1667), here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence (the eighth review's caution), and the needed ingredient is on hand: the convergent variance sum makes the partial products an L^2-bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly integrable and the limit law has mean exactly 1, not merely limit-of-means 1;* (ii) *[limit law] as $a \to \infty$ over non-cubes, $C(a)$ converges in distribution to the random Euler product $\prod_{p\equiv 1(3)} f_p(\xi_p)$ with the three-state local variables ξ_p above—independent at any finite set of primes exactly, by CRT, with the infinite product's tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 1(iii), no normalization is needed)—with derived standard deviation $0.2762\ldots$ computed from the full three-state law;* (iii) *[uniformity] $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a)I_a(N)(1 + o(1))$ uniformly over non-cube $a \leq (\log N)^B$.*

The framework here is Kowalski's theory of averages of Euler products and singular-series distributions [11], which proves such limit laws for k-tuple singular series; the one-parameter cubic shift family, its exact mean-one lemma, and the short-range uniformity appear unstated, and we present (ii) explicitly as an instance-level law inside that framework rather than a new mechanism. Verification: derived moments (mean exactly 1, sd 0.2762) against the 294 non-cube shifts $a \leq 300$: empirical mean 1.0215 (+1.5 profile standard errors), sd 0.2481 (finite-family tail correlations account for the sd gap: for $p > a_{\max}$ the residues of a are not yet equidistributed); uniformity profile over 57 shifts at $N = 2 \times 10^5$: mean $z = +0.09$, spread 0.72, max $|z| = 2.31$. (The single instance $a = 2$, $C = 1.298435 \pm 0.0003$, verified at 10^7 with $z = +0.32$, is retained as the calibration member; no statement of Bunyakovsky type is proven for any cubic, the nearest theorem being Heath-Brown's $x^3 + 2y^3$ [3], which is what keeps the whole family conjectural.)

Conjecture 3 (Contamination in prime triplets; apparently new)**.** *The contamination calculus of Conjecture 21(v) extends to k-tuples, and the triplet pattern $(n, n+2, n+6)$ is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically—(q^2, q^2+2, q^2+6) by $3 \mid q^2+2$, and (q^2-6, q^2-4, q^2) because $q^2-4 = (q-2)(q+2)$ is composite—leaving the single* doubly-thinned *survivor $(q^2-2,\ q^2,\ q^2+4)$, which requires* $q^2 - 2$ and $q^2 + 4$ *simultaneously prime and forces $q \equiv \pm 2 \pmod 5$ (else $5 \mid q^2+4$). Triplet starts lie in classes $n \equiv 1, 2 \pmod 5$, and the surviving configuration has start $q^2 - 2 \equiv 2$: class 2 is contaminated and class 1 leads,*

$$\mathcal{M}_x\left(\frac{\pi_3(\cdot;5,1) - \pi_3(\cdot;5,2) - T_3}{\sqrt{\pi_3}}\right) \to 0, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2-2,\ q^2+4 \text{ prime}}} \log(q^2-2) \log q\ \log(q^2+4),$$

with drift-to-noise $\asymp \log^{-3/2} x$—one logarithm weaker than the pair races, a scale the calculus itself predicts and the verification must respect.

This is the calculus made to pay rent at tuple level: the configuration census, the double thinning, and the mod-5 class assignment are all forced, and the prediction was derived before the data were taken. At 10^9: classes 189,837 against 189,670 ($D = +167$ on the predicted side; $T_3 = +25$ against noise 616, so the endpoint is uninformative exactly as the $\log^{-3/2}$ law requires), leadership log-density 0.65 (+0.8 null standard deviations at the corrected occupation constant—directionally consistent, sharpness unavailable at this height by the calculus's own accounting). (The Chernick chain $\{p, 2p-1, 3p-2\}$ that formerly occupied this slot—the population behind the universal-form Carmichael numbers, previously quantified by Dubner [10]—is retained as attributed calibration: $C = 2.858249$, observed 125,379 against predicted 125,429.4 at 3×10^8, $z = -0.14$.)

Conjecture 4 (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family)**.** *For $k \geq 2$ and $m \geq 2$, write $D_k(m) = m^k - (m-1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \geq 1$. Then:* (i) *[theorem] for* composite *k, no k-th power is representable: if r is a proper divisor of k with $r > 1$ (such r exists precisely because k is composite; $r = 1$ would allow the useless factor $m - j = 1$ at $j = m-1$, a gap in an earlier phrasing noted by the eighth review), then for every $1 \leq j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1;* (ii) *[elementary] for* prime *k (including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (its roots $\zeta^j/(\zeta^j - 1)$, ζ a primitive k-th root of unity, form a single Galois orbit—D_k is a linear-fractional transform of the cyclotomic polynomial Φ_k);* (iii) *[conjecture] for each prime k the representable lane follows Bateman–Horn: $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt/\log D_k(t)$.*

Finding 1. Our first version of this ladder claimed an even/odd dichotomy, with a Bateman–Horn lane for every odd k. The third external review refuted it: for odd composite $k = rs$ the factorization in (i) applies (e.g. $D_9(m)$ is *never* prime), so the lane exists only for prime k. We verified the refutation directly ($k = 9, 15, 21, 25, 27, 33$: zero prime values of D_k to $m = 2000$, with the predicted divisor $D_r(m) \mid D_k(m)$ checked identically) and record that this is the second algebraic-factorization failure the audit programme has caught in our own output—the exact failure mode our procedure was built to detect, twice missed at generation time and twice caught downstream.

The family grew out of the audit refutation behind Conjecture 13 (the case $k = 3$). It replaces, at the suggestion of an external review, a bibliographically-novel-at-most quintuplet constellation that formerly occupied this slot (its data survive in the repository; its $4 \cdot 10^9$ verification, ratio 1.0004, remains part of the audit record as Hardy–Littlewood calibration).

Conjecture 5 (Twin cyclotomic bases: the complete quadratic family; no prior trace found)**.** *For the three cyclotomic polynomials of degree 2 (indices k with $\varphi(k) = 2$, i.e. $k \in \{3, 4, 6\}$), consider $\Phi_k(n)$ and $\Phi_k(n+1)$ simultaneously prime. Then:* (i) *[parity obstruction, elementary] for $k = 4$ one member of the pair $(n^2+1,\ n^2+2n+2)$ is even for* every *n, so the only prime pair is $n = 1$, giving $(2, 5)$—the same obstruction that forbids consecutive $n^2 + 1$ primes;* (ii) *[collapse, elementary] $\Phi_6(x) = \Phi_3(x-1)$ identically, so the $k = 6$ pair at index n* is *the $k = 3$ pair at index $n - 1$: the family contains exactly one nontrivial statement;* (iii) *[conjecture, the live instance] there are infinitely many n for which the length-3 repunits in bases n and $n+1$ are simultaneously prime: $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n+1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5\, I(N)$, with $C_5 = C(x^2 + x + 1,\ x^2 + 3x + 3)$.*

Both forms in (iii) are odd for every n (each is Φ_3 of an integer), and mod 3 each vanishes on exactly one class ($n \equiv 1$ and $n \equiv 0$ respectively), so the pair is admissible with $\omega(3) = 2$. Unlike the arbitrary shifted pair that formerly occupied this slot (replaced at the suggestion of an external review), the companion here is structurally forced: it is the same cyclotomic value at the consecutive base, making this the natural "twin" statement inside the cyclotomic-chain family of Conjecture 6. The family analysis (i)–(ii) is what a family *lift* of the statement uncovers, and both clauses were found by our own admissibility machinery: the $k = 4$ branch was rejected as inadmissible at $p = 2$ at singular-series time (both-odd count to 10^6: zero, prime pairs: $n = 1$ only), and the $k = 6$ run returned counts *identical* to $k = 3$, which is the numerical shadow of the polynomial identity in (ii).

Conjecture 6 (The alternating cyclotomic chain; apparently new)**.** *There are infinitely many primes p such that, writing $u = \Phi_3(p) = p^2 + p + 1$, the three integers*

$$p, \qquad u = p^2 + p + 1, \qquad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with $\#\{p \le x\} \sim C_6^{\mathrm{ch}}\, I(x)$ for the Bateman–Horn system $\{x,\ x^2 + x + 1,\ x^4 + 2x^3 + 2x^2 + x + 1\}$, $C_6^{\mathrm{ch}} = 3.6143 \pm 0.011$. The first chains are $(2, 7, 43)$ and $(3, 13, 157)$.

This is the repunit analogue of a Cunningham chain: a prime base p, its length-3 repunit $u = (p^3 - 1)/(p - 1)$, then $(u^3 + 1)/(u + 1)$. The chain must *alternate* between Φ_3 and Φ_6: the naive iterate $\{p,\ \Phi_3(p),\ \Phi_3(\Phi_3(p))\}$ is *inadmissible*—if $p \equiv 2 \pmod 3$ then $u \equiv 1 \pmod 3$, so $3 \mid \Phi_3(u)$ always, while $p \equiv 1 \pmod 3$ forces $3 \mid u$; our admissibility check rejected the naive chain at design time. The uniqueness claim is made over a specified construction space, following external review: among length-3 chains $\{x,\ \Phi_j(x),\ \Phi_k(\Phi_j(x))\}$ with $j, k \in \{3, 6\}$ *extending the repunit sub-chain* (i.e. $j = 3$), the mod-3 arithmetic above forces $k = 6$: alternation is the unique admissible continuation. (Dropping the repunit anchor admits one sibling, the pure-Φ_6 chain $\{x, \Phi_6(x), \Phi_6(\Phi_6(x))\}$ on the complementary lane $x \equiv 1 \pmod 3$; since $\Phi_6(x) = \Phi_3(x - 1)$, its second element is again a repunit value, and we regard the classification of all admissible words in the two-letter alphabet $\{\Phi_3, \Phi_6\}$—which words are obstruction-free at every prime, and whether they form a regular language—as a natural open programme raised by the review.) The length-2 sub-chain $\{p,\ p^2 + p + 1\}$ is the classical cyclotomic Sophie Germain statement that formerly occupied this slot ($C = 1.521661 \pm 0.0005$, catalogue value $1.5217315\ldots$, OEIS A188596; it remains verified at $z = -1.06$ at 10^7 and is retained as calibration): infinitely many prime bases have prime repunits of length 3, the base-independent home of the phenomenon whose base-2 shadow is the Mersenne sequence. The length-3 chain stated here is, as far as we can find, unstated, and its quartic layer makes it the highest-degree system in the suite.

Conjecture 7 (The contamination matrix: sexy pairs with two surviving orientations; apparently new)**.** *For the pattern $(n, n + 6)$, both prime-square orientations survive, on complementary classes of q mod 5—the first matrix instance of the calculus (every pair race so far had exactly one): orientation $A = (q^2 - 6,\ q^2)$ requires $q^2 - 6$ prime, forcing $q \equiv \pm 2 \pmod 5$, and lands in start class 3 (mod 5), class 3 (mod 8); orientation $B = (q^2,\ q^2 + 6)$ requires $q^2 + 6$ prime, forcing $q \equiv \pm 1 \pmod 5$ (the term $q = 5$ falls outside the start classes), and lands in class 1 (mod 5), class 1 (mod 8). Sexy starts occupy classes $\{1, 2, 3\}$ mod 5 and $\{1, 3, 5, 7\}$ mod 8; the predicted drift vector, in the $\mathcal{M}_x$-sense of Conjecture 21 and with $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2 - 6 \text{ prime}} \log q \log(q^2 - 6)$, $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2 + 6 \text{ prime},\, q \ne 5} \log q \log(q^2 + 6)$, is: mod 5: $\pi(2) - \pi(3) = T_A$, $\pi(2) - \pi(1) = T_B$ (class 2 clean); mod 8: $\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3) = T_A$, $\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1) = T_B$ (classes 5, 7* *clean and mutually symmetric). Two independently computable drift components, one certified null class per modulus.*

The matrix structure—two orientations feeding disjoint classes on complementary q-classes, with an interior null—is what distinguishes this from every single-orientation race; the fifth review asked for exactly this test of the calculus beyond its balanced one-mechanism cases, and the class assignments were verified independently by a clean-context audit before the data were taken. At 10^9 the predicted components ($T_A = 191$, $T_B = 110$) sit far inside the noise ($\approx 2{,}100$), as the $1/\log x$ law forces; all four measured components and the 5–7 control lie within one noise unit of their predictions (trivially, at this height), and the four leadership log-densities $(0.48, 0.56, 0.53, 0.73)$ sit at $(-0.1, +0.3, +0.1, +1.2)$ null standard deviations (occupation constant 0.18): three at the null, the mod-8 B-component mildly positive on the predicted side. The matrix is *registered and consistent*, with sharpness unavailable below $\sim 10^{14}$ by the calculus's own accounting. (The pair $\{p,\ p^2-2\}$, OEIS A062326, $C = 3.383216$, formerly this slot's occupant and still the input that sets Conjecture 21's drift scale, is retained as attributed calibration: verified at 10^7, $z = -1.31$, independently recomputed to 10^9; a refutation of its infinitude would remove the twin race's predicted mechanism.)

Conjecture 8 (The null-mechanism race; apparently new)**.** *For the quadratic twin pairs of Conjecture 1 at $d = 2$, square contamination is* algebraically impossible*: $n^2+1 = q^2$ has no solution with $n \geq 1$, and $n^2+3 = q^2$ only at $(n, q) = (1, 2)$. The surviving classes of n mod 5 are $\{0, 1, 4\}$, and the classes 1 and 4 have identical singular series (elementary, via CRT). Let $\pi^{\mathrm{q}}_a(x)$ count solutions with $n \equiv a \pmod 5$, $D(x) = \pi^{\mathrm{q}}_1(x) - \pi^{\mathrm{q}}_4(x)$, and define the normalized race process $Y(x) = D(x)/\sqrt{\pi^{\mathrm{q}}_1(x) + \pi^{\mathrm{q}}_4(x)}$. Then, in contrast to the twin race of Conjecture 21, this race is* driftless*:* (i) $\mathcal{M}_x(Y) \to 0$ *(no deterministic term at the contamination scale, the negation of Conjecture 21(i) for this pair);* (ii) *[occupation law, corrected clock, stated as a conditional consequence] the arithmetic content of this clause is an* invariance principle*: the normalized race process $D(\lfloor Nt \rfloor)/\sqrt{N}$ (in event count N) converges weakly to Brownian motion. Everything after that is probability, not number theory (the eighth review's demanded separation, which we adopt): granting the invariance principle, $D(t) \approx B(N(t))$ with $N(t) \asymp t/\log^2 t$, and in* logarithmic *time $u = \log t$ the Lamperti reduction $Z(s) = \mathrm{e}^{-s/2} B(\mathrm{e}^s)$ is a stationary, ergodic Ornstein–Uhlenbeck process, so the logarithmic occupation measure of leadership converges almost surely to $\frac{1}{2}$:*

$$\frac{1}{\log x} \int_2^x \mathbf{1}_{\{D(t)>0\}} \frac{dt}{t} \ \longrightarrow\ \frac{1}{2},$$

with fluctuations of the Gaussian scale $\sqrt{\log 2/\log x}$ (the occupation-indicator covariance $\frac{1}{2\pi} \arcsin \mathrm{e}^{-u/2}$ integrates to $\frac{1}{2}\log 2$; an earlier draft used $2\sqrt{\log 2/\log x}$, which is the constant for the sign *average $2\mathbf{1} - 1$, not the occupation fraction—a factor-of-two error caught by the seventh external review, whose correction doubles every leadership significance quoted in this paper); the classical arcsine spread survives only in the* natural *event clock, i.e. for occupation fractions of $\{B(m) > 0,\ m \leq N(x)\}$. A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.*

An earlier draft of clause (ii) predicted an arcsine-distributed limit for the logarithmic occupation itself; the sixth external review pointed out that this is incompatible with the change of clock—the arcsine law lives in the natural time of the walk, and under dt/t averaging the Lamperti transform is ergodic, forcing the limit $\frac{1}{2}$. We record the correction because it changes what counts as evidence: the observed log-occupation 0.21 at 10^7 sits 1.4 null standard deviations below $\frac{1}{2}$ (scale $\sqrt{\log 2/\log x} \approx 0.21$ there)—still consistent with the driftless null—where the uncorrected clause would have read the same number as an unremarkable arcsine draw and learned nothing.

This is the negative control that completes the mechanism family of Conjectures 21 and 25: those races carry a predicted drift *because* prime squares can infiltrate specific residue classes of specific patterns; here the quadratic form blocks squares identically, so the same model predicts *nothing*—a falsifiable contrast. If a persistent drift were found in this race, the mechanism story behind the family would be wrong no matter how well the positive cases fit. One caution, prompted by external review, is built into the statement: the absence of *this* drift source does not by itself prove the race driftless—oscillatory zero terms, higher prime powers, and endpoint corrections are other conceivable sources—so clause (i) is a conjecture under the same zero-oscillation hypothesis as Conjecture 21, not a consequence of the square-blocking algebra alone. What the algebra supplies is the *contrast*: whatever residual model one adopts, it must produce drift in Conjectures 21 and 25 and none here. Verification at 10^7: 32,898 pairs, class counts 10,917 against 10,981 ($D = -64$, i.e. -0.43 noise units), log-mean normalized drift -0.46 (consistent with 0 at one noise unit), log-occupation of leadership 0.21—1.4 null standard deviations below $\frac{1}{2}$ on the corrected clock and constant of clause (ii), within the fair race's 95% band. (The quadratic triple $n^2 + \{1, 3, 7\}$ that formerly occupied this slot, $C = 10.64599 \pm 0.006$, is retired to the audit record with its calibration intact—observed 3,963 against predicted 3,998.6 at 10^7, $z = -0.56$, independently recomputed to 3×10^7; the third external review found the longer pattern $n^2 + \{1, 3, 7, 9, 13\}$ recorded in the OEIS since 2000, so the tuple family has prior presence and only its singular-series evaluation was plausibly new.)

Verification

	bound	observed	predicted	ratio	z
C1$_{d=2}$	10^7	32,898	32,862.7	1.0011	+0.19
C2$_{a=2}$	10^7	287,956	287,784.8	1.0006	+0.32
C4$_{k=2}$	10^7	1,270,606	1,270,902.8	0.9998	−0.26
C4$_{k=3}$	10^6	126,826	126,641.6	1.0015	+0.52
C4$_{k=5}$	$8 \cdot 10^5$	56,925	57,021.3	0.9983	−0.40
C5	10^7	33,274	32,976.7	1.0090	+1.64
C6	10^7	1,362	1,357.2	1.0035	+0.13

For C1 and C2, the family-profile verifications (150 and 57 shifts) are reported at the statements; the table rows are the classical single instances. The race conjectures 3 and 7 carry their data at their statements. For C4: the even-rung theorem was checked numerically to 10^8 (no fourth power is $p + j^4$); the three verifiable Bateman–Horn lanes $k = 2, 3, 5$ (constants 2 exactly, 3.36181, and 3.67770 ± 0.007, the quartic product computed by brute-force root counts to 10^5) all sit within $|z| < 0.6$. For C5, $C_5 = 2.964239 \pm 0.002$. For C6, the chain census by decade is 4, 11, 41, 224, 1362 at $N = 10^3, \ldots, 10^7$ (z from +0.40 to +0.13, no drift), the quartic layer's singular series computed by brute-force root counts to 10^5; the retained sub-chain benchmark $\{p,\, p^2 + p + 1\}$ stands at 33,661 against 33,855.5 at 10^7 ($z = -1.06$). For C8, the race data are reported at the statement. Independent clean-context recomputation gave, for the retained calibration members of slots 6, 7 and 8 (the Sophie Germain sub-chain, the pair $\{p, p^2 - 2\}$, and the quadratic triple), 1.521730, 3.383227 and 10.647706, with primes to 10^9 (each inside our stated wobble) and independent counts to 3×10^7 with final ratios 0.9962, 0.9965, 1.0059. (The quintuplet that formerly occupied slot 4 retains its audit-record calibration: observed 15,236 against predicted 15,230.3 at $4 \cdot 10^9$, ratio 1.0004, $z = +0.05$.)

4 Sparse sequences: barely-divergent Borel–Cantelli

For sequences whose nth term has size e^{cn} the expected number of primes is $\sum_n \kappa_n/(cn)$: convergence or divergence of this sum is the entire question, and when it diverges it diverges logarithmically, so the right conjecture has counting function $\alpha \log x$ with a derived α—never more.

Conjecture 9 (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327)**.** (i) *[rank-disjointness, elementary, with its exception stated] every* odd *prime factor r of L_p has rank of apparition exactly $2p$ (from $z(r) \mid 2p$, $z(r) \nmid p$, and $z(r) = 2$ being impossible), and every prime factor of F_p ($p \neq 5$ prime) has rank exactly p: the odd divisor pools of the two numbers are disjoint. The prime 2 is the unique exception: 2 divides both F_p and L_p precisely when $p = 3$ ($F_3 = 2$, $L_3 = 4$; $\gcd(F_p, L_p) = 2$ iff $3 \mid p$)—an unqualified draft of this clause was false at $p = 3$, caught by the seventh review. Disjoint pools remove shared-*prime *correlation but not all correlation: dependence can still enter through the order structure of the recurrence, which is why (iii) matters;* (ii) *[finiteness] only finitely many primes p have F_p and L_p simultaneously prime;* (iii) *[calibration clause, deliberately quantitative] the naive joint accounting—probability $(e^\gamma \log p/(p \log \phi))^2$ per index—is contradicted at the 3×10^{-3} level by the record: it prices the known index $p = 148091$ (OEIS A080327, F_p a probable prime, L_p proven) at that much expected mass beyond 10^4. A single surprising event is not a strict refutation, but it is strong evidence that the naive constant is underestimated—by roughly an order of magnitude, or through enrichment the independence model misses—and it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.*

This slot is the suite's designated stress test of the convergent Borel–Cantelli template that Conjecture 10 exemplifies, and it earns that role by *failing* the naive version in public: our scan to $p \leq 10^4$ found exactly the joint indices $\{5, 7, 11, 13, 17, 47\}$ with expected further mass 1.6×10^{-3} —and the catalogued index 148091 exists anyway. We therefore conjecture finiteness *without* a completeness clause (the Goldilocks-maximal list that is defensible for factorial twins is indefensible here, and the audit caught us before we published one); the lesson, in the language of Grantham–Granville [7], is that recurrence-sequence constants need their local corrections *before* tail masses are trusted. The retained single-sided calibration: 25 Fibonacci prime indices $\leq 10^4$ against the naive screening prediction 34.1 ($z = -1.6$, the deficit flagged since the first draft and reconfirmed independently); 29 Lucas prime indices $\leq 10^4$; the corrected c_F remains the suite's most concrete "needs a Granville-type patch" constant, and clause (iii) now quantifies the same need at the joint level.

Conjecture 10 (Factorial twins; the uniqueness core is (a), OEIS A088054; clauses (i) and (iii) are ours)**.** (i) *[window rigidity, elementary] for $n \geq 4$ and $2 \leq |a| \leq n$ the number $n! + a$ is composite (any prime factor p of a satisfies $p \leq n$, so $p \mid n! + a$, and $n! + a > n \geq p$); the two small anomalies $2! - 2 = 0$ and $3! - 3 = 3$, caught by the sixth external review against an unrestricted first draft, are the only exceptions. Hence among all offsets of bounded size the only candidates for primality near $n!$ are $a = \pm 1$: the twin question below is the* unique *bounded-offset constellation question at $n!$, and every admissible offset set is a subset of $\{-1, +1\}$.*

(ii) *[uniqueness; (a)] Only finitely many n have $n!-1$ and $n!+1$ simultaneously prime, and the complete list is $n=3$: i.e. $6=3!=3\#$ is the unique factorial lying between twin primes.* (iii) *[joint fluctuation model] the two single-sided counts $F_\pm(N)=\#\{n\le N:\ n!\pm 1 \text{ prime}\}$ are each $\sim \mathrm{e}^\gamma \log N$ and,* under the independent-indices model stated below*, asymptotically independent, with F_+-F_- normalized by $\sqrt{2\mathrm{e}^\gamma \log N}$ asymptotically standard normal—where the Gaussian limit is asserted* under the explicit summability hypothesis $\sum_{m<n\le N}\left|\mathrm{Cov}(\mathbf{1}_{E_m},\mathbf{1}_{E_n})\right|=o(\log N)$ *over the four sign pairs (the variance itself is $\asymp \log N$, so this is exactly what a dependency-graph CLT needs), not merely imported from an independence model; the dependence structure across factorial indices, i.e. whether that hypothesis holds, is the open component of this clause.*

Each event separately has Caldwell–Gallot probability $\sim \mathrm{e}^\gamma/n$ [4] and divergent sum (the $n!+1$ side is the classical law retained as calibration at Conjecture 12); the *joint* event has probability $\sim (\mathrm{e}^\gamma/n)^2$, whose sum converges. This is the suite's exemplar of the convergent side of the Borel–Cantelli dichotomy—the accounting that predicts finitude rather than infinitude. The at-a-common-index independence in (iii) has a derivation, not just an assertion (the sixth review's demand): a generic twin pair near $n!$ would carry the Hardy–Littlewood twin factor $2C_2\prod_{2<p\le n}\frac{p-2}{p-1}/\prod_{2<p\le n}\left(\frac{p-1}{p}\right)^2$-type couplings, but here the screening (coprimality to all $p\le n$) is *deterministic*—both neighbors always pass it—and the twin coupling factor and the reciprocal sieve factors cancel to first order, leaving the joint probability $\sim (\mathrm{e}^\gamma/n)^2$ with no singular-series correlation; what remains open, and is flagged in (iii), is dependence *across* distinct indices n, whose divisibility structures are nested. That open component has a concrete first computation, which we register as the slot's programme: for $m<n$ and a prime $p>n$, the residues $m! \bmod p$ and $n! \bmod p$ are deterministically linked by $n!=m!\cdot(m+1)\cdots n$, so the covariance of the events $p\mid m!\pm 1$ and $p\mid n!\pm 1$ is an explicit character-sum average over the multiplier $(m+1)\cdots n$; summing it over p either bounds the cross-index dependence at $o(1/n)$—vindicating the independent-indices model—or exhibits genuine coupling, and the computation is finite at each height. Verified for $n\le 700$ ($F_+=14$, $F_-=16$: difference -2 against noise scale ≈ 4.8); the model-expected number of further twin examples beyond 700 is $\mathrm{e}^{2\gamma}/700\approx 4.5\times 10^{-3}$ (computed tail). The defensible core is finiteness; the exact one-element list is the Goldilocks-maximal form—strictly stronger than the tail estimate can guarantee. On priority the record now reads: the fifth review asserted, and the sixth review's pointer confirmed, that the exact uniqueness conjecture is on the public record—OEIS A088054 states that 3 is (conjecturally) the intersection of A002981 and A002982—so clause (ii) is attributed there, our earlier "explicit prior statement not found" label is withdrawn (the second priority correction on this slot; Finding 2), and the slot's claimed content is exactly clauses (i) and (iii).

Finding 2. The statement that previously occupied this slot—primorial twins: $p\#\pm 1$ both prime only for $p\in\{3,5,11\}$—was carried for three revisions under an "apparently new" label, and the fourth external review found it stated in Lillie [5], whose abstract gives both the $O(n^{-2})$ joint probability and the prediction that there are in total three instances. The failure is instructive and we record it against ourselves: our novelty search had *surfaced that exact preprint*, but the search snippet described only its single-sided results, and we did not read to the abstract's second half. Priority checks on a conjecture must read the abstract and body of near-neighbour papers, not their search summaries; the audit protocol now requires it. The primorial statement remains, attributed to Lillie, as calibration (our independent verification to $p\le 4000$ stands), and the slot now carries the factorial analogue. The factorial uniqueness statement, in turn, was believed unrecorded through two further searches—until the sixth external review's assertion led us to OEIS A088054, whose comment states precisely that 3 is conjecturally the unique common index of A002981 and A002982. Our OEIS sweep had checked the two defining sequences but not the derived sequence carrying the intersection comment. The audit lesson compounds: a priority search must cover the *neighbourhood* of a statement—derived sequences, comments, and secondary results—not only its defining objects; two misses on one slot is the strongest argument in this paper for that protocol.

Conjecture 11 ($n^2 + 2^n$; sequence is OEIS A064539)**.** *Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod 6$ (elementary), and:* (i) *infinitely many such primes exist, with counting function $\asymp \log N$;* (ii) *[candidate law, entanglement-aware form] for a finite set S of primes $p \geq 5$ (the primes 2 and 3 are exact in the lane reduction, $\delta_2 = \delta_3 = 0$, and $\operatorname{ord}_2 2$ is undefined—the sixth review's point) let D_S be the* exact *joint density, within the lane $n \equiv 3$ (6), of n with $p \nmid n^2 + 2^n$ for all $p \in S$, computed over the full joint period $\operatorname{lcm}_{p \in S} \operatorname{lcm}(6, p \cdot \operatorname{ord}_p 2)$, and set $\kappa_S = D_S / \prod_{p \in S}(1 - 1/p)$. The conjecture is stated along the* canonical exhaustion *$S_z = \{p \text{ prime} : 5 \leq p \leq z\}$, and its three layers are logically separate and are asserted separately (the eighth review's demanded separation):* (ii-a) *the sequence κ_{S_z} converges as $z \to \infty$ (a general net over arbitrary finite S carries an order-of-exhaustion ambiguity we do not need), with limit κ;* (ii-b) *whether the limit* factors*—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is a separate question, on which (ii-a) takes no position; and* (ii-c) *the count is $\sim \sum_{n \leq N,\, n \equiv 3\,(6)} \kappa / \log(n^2 + 2^n)$—a genuine further step, since a convergent local density does not automatically govern the global count. The working value $\kappa^* = 4.2734\ldots$ is explicitly a* hybrid*, not a computation of any single κ_{S_z}: the CRT-exact core S_{19} times independent per-prime factors for $19 < p \leq 300$—so $\kappa^* = \kappa_{S_{300}}$ exactly if and only if the observed exact factorization persists to* 300*.*

This is not a Bateman–Horn problem: the local conditions at different primes are tied through the orders $\operatorname{ord}_p 2$, and the per-prime product $\prod_p (1 - \delta_p)/(1 - 1/p)$ of an earlier draft tacitly assumed their independence. Following external review, the law is now stated through the CRT-exact quantities κ_S, which make no independence assumption. The computation then returned a small surprise in the other direction: through $p \leq 19$ (joint period 116,396,280) the joint survivor fraction equals the product of the per-prime fractions as an *exact rational identity*, verified in exact arithmetic at every level $S = \{5 \leq p \leq P\}$, $P \in \{5, 7, 11, 13, 17, 19\}$ (the sixth review rightly objected to the earlier "machine precision" phrasing; the identity is exact, not approximate). The entanglement the reviewer warned about is thus absent at all computed levels; whether exact factorization persists for all p—equivalently, whether the survival events are exactly independent over every joint period, which would itself be a striking equidistribution statement about the orbits of 2—is part of what clause (ii) asks. We separate the robust claim (i) from the candidate law (ii), and note that the eight hits below 6000 are far too few to validate a four-digit constant—the verification below checks consistency, not the constant. The sixth review also reports OEIS-recorded candidate indices beyond our completed range: 29355, 34653, 57285, 99069, and the probable-prime candidate 1933695. We could not independently re-verify these (and label them accordingly), but all five satisfy the lane constraint $n \equiv 3$ (6), and taking them at face value the κ^*-model predicts 2.9 hits in $(6 \times 10^3, 10^5]$ against 4 reported and 5.9 in $(6 \times 10^3, 1.94 \times 10^6]$ against 5 ($z = +0.65$ and -0.38): the records extend the consistency check by two decades.

The congruence claim is elementary: n odd is forced by parity, and for odd n with $3 \nmid n$ we have $n^2 + 2^n \equiv 1 + 2 \equiv 0 \pmod 3$. What appears to be a harsh obstruction is, inside the surviving lane, a local bonus: $\delta_2 = \delta_3 = 0$ contribute the factors $2 \cdot \frac{3}{2} = 3$ to κ. Mixed polynomial–exponential sequences are exactly where hand-waving fails and the computed local product earns its keep.

Conjecture 12 (The pair-level Montgomery–Soundararajan reduction; apparently new)**.** *For the window field of Conjecture 16(ii), the diagonal variance of the twin-pair count reduces to a* pinned *singular-series average. With $R(h) = C_4(0,2,h,h+2)/\mathfrak{S}(2)^2$ for $h \geq 1$ (zero when the offset set is degenerate or inadmissible—all odd h, and $h = 2$ since $\{0,2,4\}$ dies at 3; consecutive twin pairs cannot overlap beyond $(3,5,7)$, so no overlap term exists) and*

$$G(H) \;=\; \sum_{1\leq h\leq H}\Big(1-\frac{h}{H}\Big)\big(R(h)-1\big),$$

the derived identity is

$$\frac{\mathrm{Var}_t\ \pi_2(t;H)}{\mathbb{E}_t\,\pi_2(t;H)} \;=\; 1+\frac{2\,\mathfrak{S}(2)\,G(H)}{\log^2 x}+o\Big(\frac{1}{\log^2 x}\Big):$$

the entire diagonal deficit left open at Conjecture 16(ii) is the single computable function G—the pair analogue of Montgomery–Soundararajan's reduction of the prime-count variance to $\sum_{h\leq H}\mathfrak{S}(h)(H-h)$ [19]. Clause (i) is a conditional proposition*, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii): $-G(H)/\log H$ diverges —the pinned average grows strictly faster than the single-prime secondary term $\frac{1}{2}(\log H+\gamma+\log 2\pi-1)$, so pair counts in windows are more sub-Poisson than prime counts—with the honest caveat, per the seventh review, that data to $H = 3000$ cannot distinguish divergence from a pure* log *with a large constant (nor $c\log H\log\log H$ from $(\log H)^\alpha$): what the computation* establishes *is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.*

G was computed exactly to $H = 3000$ (4-tuple constants by Euler products to 2×10^4): $G(3000) = -20.9$, with local logarithmic slope increasing from 3.4 to 3.7 across the range—seven times the single-prime slope $\frac{1}{2}$, consistent with (ii) over the computable range while leaving the limiting form genuinely undetermined (a quadratic-in-log fit is unstable on subranges). The window measurement brackets the extrapolation: at $H = 10^5$, $x \in [10^8, 2\times 10^8]$ the pure-log form predicts $\mathrm{Var}/\mathbb{E} = 0.97$, the $\log^2$ form 0.72, and the observed value is 0.815—intermediate growth, exactly where the computable range points. (Caldwell–Gallot's factorial-prime law $\#\{n \leq N : n!+1 \text{ prime}\} \sim \mathrm{e}^\gamma \log N$ [4], this slot's former occupant and the divergent single-sided companion of Conjecture 10, is retained as attributed calibration: 15 primes to $n = 700$ against 11.7, $z = +0.98$.)

Verification

C9: joint scan and single-sided calibration reported at the statement (Fibonacci prime indices $\{3,5,7,\ldots,9311,9677\}$, 25 of predicted 34.1, $z = -1.56$; Lucas indices 29; joint $\{5,7,11,13,17,47\}$ to 10^4). C10: exhaustive to $n \leq 700$: the only factorial twin is $n = 3$; individually, $n!+1$ is prime for $n \in \{2,3,11,27,37,41,73,77,116,154,320,340,399,427\}$ and $n!-1$ for $n \in \{3,4,6,7,12,14,30,32,33,38,94,166,324,379,469,546\}$—two divergent single-sided laws straddling their shared constant while the joint count stops at one, the convergent accounting in action. (The primorial companion, now attributed to Lillie [5] per Finding 2: exhaustive to $p \leq 4000$, twins exactly $\{3, 5, 11\}$, with $p\# + 1$ prime for eleven p against the Caldwell–Gallot prediction 14.8, $z = -0.98$.) C11: hits $\{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up to 6000: observed 8, model 8.55, $z = -0.19$ (at the earlier bound 4200: 8 vs 8.19, $z = -0.07$); the lane constraint was verified exhaustively (no prime off $n \equiv 3$ (6) to $n = 3000$, independently reconfirmed), and the CRT-exact factorization of clause (ii) was verified through $p \le 19$ as described at the statement. C12: the pinned singular-series computation and window comparison are reported at the statement; its retained factorial benchmark stands at 15 primes with $n \le 700$ against 11.7 ($z = +0.98$), the divergent single-sided companion of Conjecture 10's convergent side.

5 Representation problems

Theorem 1 (Cube obstruction; found by the adversarial audit)**.** *For $k \ge 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \ge 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable* cubes *alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 13(i), and is conjectural.)*

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \le j \le k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \le K$, by an upper-bound sieve of Brun or Selberg type applied to the quadratic's prime values), all but a vanishing proportion of cubes are unrepresentable. □

Finding 3. As first generated ("the exception set of $n = p + k^3$, $k \ge 1$, is finite"), the next conjecture was **false**, and our own adversarial battery refuted it: all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes, explained by Theorem 1. The error is a textbook violation of the admissibility clause of the design criteria—an algebraic obstruction along a density-zero subsequence, invisible to a Borel–Cantelli sum taken over all n. We record the failure rather than hide it; the restated version separates theorem from conjecture.

Conjecture 13 (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham's cuban observation (OEIS A002407); the family classification is apparently new)**.** *For $F \in \mathbb{Z}[x]$ with* $\deg F \ge 2$ *and* positive leading coefficient*, and $m > j \ge 1$, $(m - j) \mid F(m) - F(j)$ (textbook), and the divided difference $(F(m) - F(j))/(m - j)$—whose leading form is then positive on the cone $m > j \ge 1$—exceeds* 1 *outside an effectively bounded region; so $F(m) - F(j)$ prime forces $m - j = 1$ apart from finitely many exceptional pairs (for the family below the exceptional set is provably* empty*). Both hypotheses are necessary: for* $\deg F = 1$ *the cofactor can be identically* 1 *and the collapse fails (an omission caught by the seventh review), and for negative leading coefficient the divided difference tends to $-\infty$ on the cone, so the "exceeds* 1*" clause is false as stated (caught by the eighth review; one may equivalently keep general sign and read the claim through $|F(m) - F(j)|$). Every representation problem $F(m) = p + F(j)$ in this range thus collapses to the boundary polynomial $D_F(m) = F(m) - F(m - 1)$. The content is the classification of the boundary lanes. For $F = x^3 + cx$, $c \ge 0$:* (i) *[trichotomy, elementary] $D_F = 3m^2 - 3m + 1 + c$ and the lane is* dead-parity *for c odd (D_F always even),* dead-3-adic *for even $c \equiv 2 \pmod 3$ ($3 \mid D_F$ always), and admissible exactly for $c \equiv 0, 4 \pmod 6$;* (ii) *[uniform boundary law] over the admissible lanes, $\#\{m \le M : D_F(m) \text{ prime}\} \sim C(D_F)\, I(M)$ uniformly for $c \le (\log M)^B$—a Bateman–Horn family whose members are indexed by the ladder* *classification rather than chosen ad hoc;* (iii) *[attributed core] the case $c = 0$ is the cuban ladder: Cunningham's classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of $n = p + k^3$ finite (Hardy–Littlewood's $E_3(X) = O(1)$ [1]).*

The reducible-boundary phenomenon is part of the same classification: $F = x^4$ has $D_F = (2m-1)(2m^2-2m+1)$—the boundary lane itself factors, which is precisely the composite-k obstruction of Conjecture 4 seen from the ladder side, and $F = x^3 + x^2$ gives $D_F = m(3m-1)$, dead by reducibility. Verified: the collapse is algebraically empty for $x^3 + cx$ ($c \geq 0$), the trichotomy checked to $m = 5000$ on all $c \leq 12$, and the two smallest admissible new lanes counted at $M = 10^6$: $c = 4$ ($C = 2.12956$, $z = +0.27$) and $c = 6$ ($C = 2.68954$, $z = -0.92$). (Empirically the non-cube exception census of (iii) by decade is 4, 27, 168, 763, 2011, 2808, 1181, 88 up to 10^8, peaking at the sixth decade and collapsing thereafter, largest found 78,526,384; verification data, not statement.)

Conjecture 14 (The Stern lane race; apparently new)**.** *In Stern's representation problem $n = p + 2k^2$ (n odd), race the k-even and k-odd lanes. Prime-square contamination of the Λ-weighted lane counts comes from $n = q^2 + 2k^2$—the norm form $x^2 + 2y^2$ of $\mathbb{Q}(\sqrt{-2})$ with prime x—and since $q^2 \equiv 1 \pmod 8$ while $2k^2 \equiv 0$ or $2 \pmod 8$ by the parity of k:* (i) *[class assignment, direct congruence] contamination sits in the k-*even *lane iff $n \equiv 1 \pmod 8$, in the k-*odd *lane iff $n \equiv 3 \pmod 8$, and for $n \equiv 5, 7 \pmod 8$* no *square contamination exists at all: $q^2 + 2k^2 \equiv 1$ or $3 \pmod 8$ always, a two-line congruence (an earlier draft called this "genus theory," which the seventh review rightly deflated—genus theory enters only for the converse question of which n* are *represented), so the null classes are provable, not conjectured;* (ii) *[drift law] with $R_\mathrm{e}, R_\mathrm{o}$ the lane counts and D the clean-minus-contaminated difference, $D(n) = D_\mathrm{sys}(n) +$ noise in the log-sampled mean, where $D_\mathrm{sys}(n) = \big[\sum_{(q,k):\, q^2+2k^2=n,\ q \text{ prime}} \log q\big]/\lambda(n)$—the weight is $\Lambda(q^2) = \log q$, each representation counted once in the $k \geq 1$ convention (an earlier draft doubled this, a normalization error caught by the seventh review; the ordered-representation factor 2 belongs to Conjecture 25, not here)—with $\lambda(n)$ the lane mean of $\log p$; on the null classes the lane difference has zero drift.*

This is the calculus applied to a *representation* problem with norm-form contamination—the drift weight is carried by the arithmetic of $\mathbb{Q}(\sqrt{-2})$ rather than by a polynomial lane—and it comes with two provable null classes, the strongest internal controls in the mechanism family. One honest caveat on depth, prompted by the eighth review: an *asymptotic* for the contaminating count $\#\{(q,k) : q^2 + 2k^2 = n,\ q \text{ prime}\}$, averaged over n, is a norm-form-with-prime-argument problem of genuine analytic difficulty (the same species as primes represented with restricted variables), and nothing here pretends otherwise: $D_\mathrm{sys}(n)$ is *computed exactly per sample*, so the verification tests the transfer conjecture against the true census, not against an unproved averaged asymptotic. Verified on 2,500 log-sampled odd $n \leq 10^8$ (mean $\sim$330 representations each): contaminated strata pooled $D = +0.27 \pm 0.33$ against predicted $D_\mathrm{sys} = +0.15$ (at the corrected $\Lambda(q^2)$ weight); null strata pooled -0.22 ± 0.33, consistent with zero (one of the four strata sits at -2σ individually, within a four-test family). (Stern's exceptional list —exactly ten odd integers are not $p + 2k^2$, the largest 5993, OEIS A060003, with the seven prime members the Stern primes—is retained as the attributed benchmark that anchors this slot: verified here to 10^9 with no eleventh exception.)

Conjecture 15 (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated)**.** *Let $s(n)$ be the least prime $p \nmid n$ with $n - p$ prime, and time-change by the expected-arrivals clock $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n),\, p \nmid n} 1/\log(n-p)$.*

Then: The ensemble is fixed first (the seventh review is right that $U(n)$ is deterministic at each n, so an expectation needs a declared measure): for a scale X,

$$\mathbb{E}_X U = \frac{\sum_{X<n\le 2X,\ 2|n} U(n)/n}{\sum_{X<n\le 2X,\ 2|n} 1/n}, \qquad \Theta_G(X) = (1-\mathbb{E}_X U)\log X.$$

(i) *[limit law] under $\mathbb{E}_X$-sampling, $U \Rightarrow \mathrm{Exp}(1)$ as $X\to\infty$;* (ii) *[canonical ordering deficit] $\Theta_G(X)\to\Theta_G>0$—the Goldbach sibling of the least-prime deficit of Conjecture 22(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes* early *and $\mathbb{E}_X[U]<1$.*

The extremal theory of $s(n)$ is well developed—Granville, van de Lune and te Riele conjectured $s(n) = O(\log^2 n \log\log n)$, and Oliveira e Silva–Herzog–Pardi [12] tabulate first occurrences of minimal Goldbach summands to 4×10^{18} against prime k-tuple predictions—but the time-changed distributional law (i) and the deficit constant (ii) appear unstated (we could not rule out that the untransformed version of (i) is implicit in the comparisons of [12], and say so). Measured on 2,000 log-sampled $n\le 10^8$: $\mathbb{E}[U] = 0.796$, with $\Theta_G = 3.12\pm 0.35$ on $[10^6,10^7)$ and 3.42 ± 0.45 on $[10^7,10^8)$—positive, stable, and roughly *twice* the least-prime-in-progressions deficit $\Theta = 1.67$ of Conjecture 22, a comparison the occupancy expansion there should eventually explain; the Kolmogorov–Smirnov distance to $\mathrm{Exp}(1)$ (0.088 at this size) is dominated by the deficit displacement itself, as (ii) requires. The derivation route for Θ_G, parallel to the occupancy expansion at Conjecture 22: expand the survival probability $\Pr(s(n)>y) = 1-\sum_{p\le y} h_p + \sum_{p_1<p_2\le y} h_{p_1,p_2} - \cdots$, where the pair term requires $n-p_1$ and $n-p_2$ simultaneously prime and so carries the Hardy–Littlewood singular series of the shift p_2-p_1; inserting the k-tuple predictions and integrating against the time change isolates the $1/\log X$ coefficient—deriving Θ_G rather than measuring it is this slot's stated programme, and the seventh review is right that $\mathfrak{S}(n)$ alone does not encode the between-candidate dependence the pair term carries. (The $(3,3)$-lane refined Goldbach benchmark formerly here—every $n\equiv 2\ (4)$, $n\ge 6$, is $p+q$ with $p\equiv q\equiv 3\ (4)$, with the halved singular-series count law, previously formulated by K. Martin [13]—is retained as attributed calibration: exhaustively verified to 10^9, count formula matching to 0.2–1.3%; its comparative lane statement is Conjecture 25.)

Conjecture 16 (Uniform quantitative de Polignac; (a)-core)**.** (i) *$\pi_d(x) = \#\{p\le x-d : p,\ p+d \text{ prime}\} = \mathfrak{S}(d)\,\mathrm{Li}_2(x)\,(1+o(1))$ uniformly for even $d\le(\log x)^A$.* (ii) *[residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even $d\ne d'$ define the overlap constant*

$$K(d,d') = C_3(0,d,d') + C_3(0,d',d+d') + C_3(0,d,d+d') + C_3(0,|d-d'|,\max(d,d')),$$

where C_3 denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). The field is randomized canonically, following external review (covariances of the deterministic cumulative counts are otherwise undefined): for t uniform on $[x,2x]$ and a window length H, let $\pi_d(t;H) = \#\{t<n\le t+H : n,\ n+d \text{ prime}\}$ and let $Z_d(t;H)$ be its studentization. The mean count per window is $\asymp H/\log^2 x$, and the regime is set by that ratio (a phase condition the sixth external review supplied against an earlier draft's bare $H\to\infty$):

$$\mathrm{Cov}_t(\pi_d,\pi_{d'}) \sim \frac{K(d,d')\,H}{\log^3 x}, \qquad \rho(d,d') \sim \frac{K(d,d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')}\,\log x},$$

where the shared-prime (triple) term is the first *covariance contribution but not the whole of it: the full expansion, per the seventh review, is*

$$\mathrm{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d,d')\,H}{\log^3 x} + \frac{1}{\log^4 x}\sum_{|h|\le H}(H-|h|)\,K_4(d,d';h) + \mathit{error},$$

with K_4 the centered singular series of the off-index four-point sets $\{0,d,h,h+d'\}$—the cross analogue of the pinned diagonal average of Conjecture 12—where, per the eighth review's caution, degenerate offset relations are handled inside the kernels, not ignored: whenever the four-point set collapses (repeated entries, as at $h=0,\ h=d,\ h=-d'$, or $d'-d=\pm h$) the configuration is a triple or pair and its contribution belongs to the K-term (or to the mean), so K_4 is defined as the centered singular series of genuinely four-element *sets, with special pairs such as $d'=2d$ acquiring their extra overlap orientations in $K(d,d')$'s census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies $\sum(1-|h|/H)K_4 = o(\log x)$, i.e. in a range of H* narrower *than $x^{o(1)}$ once the Conjecture-12 growth of such sums is taken seriously; the conjecture asserts the displayed two-term form with the dominance range as part of the statement. When $H/\log^2 x \to \lambda$ the window counts converge jointly to Poisson laws with these vanishing covariances, and when $H/\log^2 x \to \infty$ every fixed finite collection $\{Z_{d_1},\dots,Z_{d_r}\}$ is asymptotically jointly normal with* independent *limiting coordinates: both kernels are finite-x corrections to independence (test below), not nondegenerate limiting kernels. The diagonal correction to $\mathrm{Var}_t(\pi_d)$ is Conjecture 12's pinned average, stated there rather than folded into a fitted σ.*

The kernel is verified twice over. Direct window-field measurement ($x=10^8$, $H=10^5$, 2000 windows, all even $d\le 40$, 190 pairs): the empirical correlation matrix matches the predicted kernel entrywise with correlation 0.86, mean off-diagonal 0.33 observed against 0.37 predicted, the small deficit having the sign of the omitted diagonal corrections, which are also directly visible as $\mathrm{Var}/\mathrm{mean} = 0.94$ on average across d. Second, the deterministic cumulative profile $z_d(x) = (\pi_d(x) - \mathfrak{S}(d)\,\mathrm{Li}_2(x))/\sqrt{\mathfrak{S}(d)\,\mathrm{Li}_2(x)}$—the estimator the uniformity verification in (i) uses—shows the same structure at cumulative scale (there $\rho \sim K\,\mathrm{Li}_3/\sqrt{\mathfrak{S}\mathfrak{S}'}\,\mathrm{Li}_2$): mean off-diagonal $\rho = 0.43$ over all even $d,d'\le 60$ at $x=10^8$, hence predicted profile spread $\sigma^2 \le 1-0.41 = 0.59$ before the diagonal correction, against observed ≈0.27, implying a diagonal factor ≈0.5 of Montgomery–Soundararajan sign and order. An earlier draft asserted only "$\sigma^2\le 1$, determined by the covariance structure"; the fourth review rightly called that a task rather than a law, the kernel above is the law the task produces, and the fifth review's demand for a genuine probability space is met by the moving-window field.

The force of the statement is its uniformity: the singular series takes a thousand different values over $d\le 2000$, spanning a factor of about three between $d=2$ and the primorial-rich d, and the counts must reproduce the entire profile simultaneously with square-root errors. A conspiracy would need to bend a thousand-dimensional vector, not one number. At $x=10^8$: correlation 0.999999, slope 0.99966, $\max_d |z_d| = 1.80$; stressed to $d\le 6000$, $\max|z_d| = 2.01$ over three thousand values.

Conjecture 17 (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner's [9]; the orientation decomposition apparently new)**.** *For even n let $R_T(n)$ count ordered (a,b), $a+b=n$, with a and b both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in $t=a$ with* *root set $\{0, \pm 2\} \cup \{n, n \pm 2\}$-type mod each p, hence its own n-dependent singular series $\mathfrak{S}_4^{(o)}(n)$, and*

$$R_T(n) \;=\; \Big[\sum_o \mathfrak{S}_4^{(o)}(n)\Big] \int_5^{n-5} \frac{dt}{\log^2 t \, \log^2(n-t)} \, (1 + o(1)),$$

the four constants computed exactly per n from the coincidence pattern of the root sets (which of $n, n\pm 2, n\pm 4$ each small prime divides). The error term is stated as $(1+o(1))$, not $(1+O(1/\log n))$: our own measurements show a second-order deficit whose $1/\log n$ coefficient is not yet pinned down (below), and claiming the rate before identifying the coefficient would outrun the evidence.

Dubner's paper already contains the integral kernel (his quotient μ_4) and the qualitative basis conjecture; the content claimed here is exactly the orientation-resolved singular series and its n-profile. The four orientations are not unrelated, and two exact identities compress them: the substitution $t \mapsto n - t$ exchanges the two summand roles, forcing $\mathfrak{S}_4^{(\mathrm{lu})}(n) = \mathfrak{S}_4^{(\mathrm{ul})}(n)$ identically (each of the (ll) and (uu) systems being self-dual under it), and the affine substitution $t \mapsto t + 2$ gives $\mathfrak{S}_4^{(\mathrm{uu})}(n) = \mathfrak{S}_4^{(\mathrm{ll})}(n - 4)$. So only one function of n per symmetry class is free, and the profile is really two-dimensional; determining the average of $\sum_o \mathfrak{S}_4^{(o)}(n)$ over n in closed form is the slot's registered identity-hunting programme (the eighth review's suggestion, which we adopt). Verified on 150 log-sampled $n \le 10^8$: the profile *shape* is confirmed at log-log correlation 0.9993 across two decades, while the *level* runs at 0.87 of prediction on $[10^6, 10^7)$ and 0.81 on $[10^7, 10^8)$—a systematic deficit of second-order size ($\approx 3.4/\log n$ at the top range, the same order as the 18–28% finite-height Poisson deficit of Conjecture 20), which we flag as the slot's open component rather than absorb into a fitted constant: either the 4-tuple second-order correction accounts for it, or the orientation model needs repair, and the two are distinguishable at 10^{10}. (The attributed benchmark stands inside the statement: every even $n \ge 4210$ is a sum of two twin members, the 35 exceptions of OEIS A007534 complete—verified to 10^8, adversarially re-swept to 10^9, independently re-verified by a different algorithm, same 35 exceptions each time.)

6 Statistical laws

Conjecture 18 (The sub-diffusivity of balanced prime races; apparently new)**.** *Model a balanced race (Conjecture 21(ii)'s symmetric class differences) as the walk of its class-assignment steps. Then the walk is measurably* sub-diffusive*:* (i) *[decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [24], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law*

$$\rho_k(x) \;=\; -\,\frac{c_k \log\log x + d_k}{\log x}\,(1 + o(1)),$$

with constants c_k, d_k in principle derivable from the LOS correlation series (underived here; the seventh review is right that the measured values $\rho_1 \approx -0.037$, $\rho_2 \approx -0.009$ at 10^9—universal across twin and cousin races mod 5 and 8 to within 0.004, each ~50 standard errors from zero—are observations at one height, consistent with $c_1 \approx 0.25$ at that height, not asymptotic constants); (ii) *[long memory, measured] short lags do not exhaust the deficit: the running maxima $M(x) = \max_{t \le x} |D(t)|/\sqrt{count}$ of the four races land at quantiles 0.02–0.17 of the simulated independent-step null (median 2.34 at this span), and still only 0.03–0.23 after correcting for $\rho_{1,2}$: the negative correlation persists across lags, so the cumulative diffusivity $\sigma^2(x) = 1 + 2\sum_k \rho_k(x)$*

sits well below 1 *at this height. The Green–Kubo sum requires more than the fixed-lag law of (i), and the eighth review is right to insist on the gap: the per-lag decay controls each $\rho_k(x)$ but not the sum unless the decay is* uniform in k with summable tail*, since the number of relevant lags may grow with x. We therefore state the needed hypothesis as part of the clause: $\rho_k(x)$ obeys the law of (i) uniformly for $k \le K(x)$ with $\sum_{k>K(x)} |\rho_k(x)| = o(\log\log x/\log x)$ for some $K(x) \to \infty$—the measured correlation length at 10^9 is consistent with K of a few tens—under which $\sigma^2(x)$ returns to* 1 *asymptotically, making sub-diffusivity a finite-height phenomenon of* $\log\log x/\log x$ *size, like every other second-order law in this paper;* (iii) *[registered dichotomy, open] asymptotically, either the races are genuinely diffusive with $\sigma^2(x) \to \sigma_\infty^2 > 0$ (running maxima $\sim \sigma_\infty\sqrt{2\log\log x}$, Darling–Erdős class), or they inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical races (maxima $\asymp$ $(\log\log\log x)^A$, the Montgomery–Ng class); no explicit formula is known for twin patterns, the two classes differ only beyond any computable height, and we register the dichotomy without choosing—what is conjectured is the finite-height law (i)–(ii).*

The measurement is the content: the repulsion constants ρ_1, ρ_2 are pattern-independent across our four races—a universality the LOS correlation series should predict quantitatively, and the slot's stated derivation programme—and the suppressed maxima are the first (to our knowledge) direct measurement of race sub-diffusivity, an effect invisible to endpoint statistics. (The twin-gap record benchmark formerly here—$G_t(x) \asymp \log^3 x$ with working constant $1/(2C_2)$, previously in Kourbatov's extreme-value framework [15]—is retained as attributed calibration: records to $4 \cdot 10^9$ wander in $[0.41, 0.56] \log^3 x$, approaching from below on a $\log\log$ clock, largest observed twin-to-twin gap 5292 after the twin at 2,466,641,069; the Cramér–Granville fragility of such constants [17] is why the constant was always flagged as first-order.)

Conjecture 19 (First occurrences of prime gaps, liminf form; (b))**.** *Let $\mathcal{G}$ be the set of even g that occur as gaps between consecutive primes (all even g, under Polignac's conjecture), and for $g \in \mathcal{G}$ let $p(g)$ be the prime starting the first such gap. Then:* (i) $\log p(g)/\sqrt{g}$ *is bounded between positive constants on $\mathcal{G}$;* (ii) *[liminf law, stated as the dual of the maximal-gap limsup,* conditional on the realization hypothesis below*]*

$$\liminf_{\substack{g\to\infty \\ g\in\mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} \;=\; \sqrt{\mathrm{e}^{\gamma}/2} \;=\; 0.943682\ldots$$

The inversion of Granville's $\limsup G(X)/\log^2 X = 2\mathrm{e}^{-\gamma}$ *into first-occurrence coordinates is* not *automatic, and we do not pretend it is: the limsup controls how large maximal gaps get, not which* gap *sizes* are *realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the* realization hypothesis*: infinitely many $g \in \mathcal{G}$ first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with $g = (2\mathrm{e}^{-\gamma} + o(1)) \log^2 p(g)$. Under that hypothesis the displayed value is forced by algebra; without it only the lower bound* $\liminf \log p(g)/\sqrt{g} \ge \sqrt{\mathrm{e}^{\gamma}/2}$ *follows from the envelope. The value is registered* as *that dual (against the slope-1 laws), not claimed as an independent phenomenon;* (iii) *[second-order, singular-series dependence] sampling g uniformly from the realized gaps $\mathcal{G} \cap [G, 2G]$ and letting $G \to \infty$,*

$$\log p(g) \;=\; \sqrt{g} + \tfrac{1}{2}\log g - \tfrac{1}{2}\log \mathfrak{S}^*(g) + E_g, \qquad \mathfrak{S}^*(g) = \prod_{\substack{p\mid g \\ p>2}} \frac{p-1}{p-2},$$

where the error E_g converges in distribution: for the sampled g,

$$\Pr[E_g \le t] \;\longrightarrow\; 1 - \exp(-\mathrm{e}^{2(t-\mu)}) \qquad (G \to \infty)$$

for some centering constant μ—the min-type (reversed) Gumbel law at scale $\frac{1}{2}$, *i.e. the law of $\frac{1}{2}\log W$ for W a unit exponential. The scale is forced by the hazard computation and is not a free parameter: the cumulative intensity of gap-g events is $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2)\exp(y - g/y)$ in $y = \log x$ (the factor* $\mathrm{e}^{-g/y}$ *is the no-intervening-prime probability), so $\frac{d}{dy}\log\Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$ at the centering point $y_0 \approx \sqrt{g}$, and the first-passage identity $\Lambda_g(y) = W$ gives $E_g = y - y_0 = \frac{1}{2}\log W + o(1)$—an earlier draft wrote the law at scale 1, overlooking that the slope of the log-intensity at y_0 is 2, not 1 (both the y-term and the g/y-term contribute); the eighth external review caught this, and the same slope-2 expansion is what produces the $\frac{1}{2}$ coefficients of the centering. (Gumbel-type modeling of gap extremes and first occurrences is Kourbatov–Wolf territory [15, 25]; the claim of this clause is only the $\mathfrak{S}^*$-centering and the scale-$\frac{1}{2}$ first-passage form, not the extreme-value framework.) Smooth gaps, whose tuple constant is large, appear* earlier *by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient $-\frac{1}{2}$, not a free parameter, while μ is left as the one unresolved constant of the clause. (The restriction to $\mathcal{G}$ repairs a definitional defect caught by external review: $p(g)$ is not known to be defined for every even g, and quantifying over all g would smuggle Polignac's conjecture into the statement. The $\mathfrak{S}^*$-clause answers the fifth review's observation that the Hardy–Littlewood factor of g materially shifts waiting times; it does not move the liminf constant, since* $\log\mathfrak{S}^*(g) = O(\log\log g)$.*)*

Clause (iii) was tested on the full first-occurrence table to 10^9: over the 100 even gaps realized in $[60, 500]$, regressing $\log p(g) - \sqrt{g} - \frac{1}{2}\log g$ on $\log\mathfrak{S}^*(g)$ (with a $\sqrt{g}$ trend term absorbing the liminf-versus-typical drift) gives coefficient -0.466 against the predicted $-\frac{1}{2}$, partial correlation -0.24: the singular series of the gap is visible in when the gap first appears, at the predicted strength.

An earlier draft asserted the full limit, inverting Granville's $\limsup G(x)/\log^2 x = 2\mathrm{e}^{-\gamma}$; an external review correctly observed that this inversion is invalid. A limsup envelope for maximal gaps controls only where gaps *can first appear at the earliest*, i.e. a lower envelope for $p(g)$—hence a liminf claim, not a limit. Whether $\log p(g)/\sqrt{g}$ converges at all, and if so whether the limit is Shanks's classical 1 [16] (as Wolf's refinement $p(g) \sim \sqrt{g}\,\mathrm{e}^{\sqrt{g}}$ and the Kourbatov–Wolf residue-class laws [25] also assert) or the envelope value, involves at least three distinct phenomena—whether a given g occurs near the extreme scale at all, the distribution of exact gap lengths near records, and the difference between record gaps and non-record first occurrences—and is left open here. The two candidate constants differ by 6%, beneath the resolution of any feasible computation: our regression slope over $g \ge 100$ falls from 1.307 at 10^9 to 1.297 at $4 \cdot 10^9$, far above both and decreasing, as it must be for either.

Conjecture 20 (Microscopic variance law; (b); the constant is Montgomery–Soundararajan's [19])**.** *Fix $\lambda > 0$ and $c \in (0, 1]$, and let X count primes in $[t, t + \lambda\log x)$ for t uniform on $[x, (1+c)x]$. Then, under the Hardy–Littlewood k-tuple conjectures, $X \Rightarrow$* Poisson(λ) *(Gallagher's argument [18], which is conditional on exactly those conjectures), and—*conditional on the quantitative hypothesis *spelled out after the statement (uniform Hardy–Littlewood with $o(h)$ control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite x*

$$\frac{\operatorname{Var} X}{\mathbb{E}\, X} \;=\; 1 - \frac{\log(\lambda\log x) + \gamma + \log 2\pi - 1}{\log x} + o\Big(\frac{1}{\log x}\Big),$$

with $\gamma+\log 2\pi-1 = 1.41509\ldots$, the leading correction being independent of the sampling constant c. [Interpolation clause] More generally, for windows of length $H=\lambda(\log x)^\alpha$ the same *formula with $\log H$ in place of $\log(\lambda\log x)$,*

$$\frac{\operatorname{Var} X}{\mathbb{E}\, X} \;=\; 1-\frac{\log H+\gamma+\log 2\pi-1}{\log x}+o\Big(\frac{1}{\log x}\Big),$$

holds uniformly for $1\le\alpha\le A$ (any fixed A with $H\le x^{o(1)}$): one expression interpolates from Gallagher's Poisson boundary ($\alpha=1$, deficit $\asymp \log\log x/\log x$) through the mesoscopic Montgomery–Soundararajan regime (deficit $\asymp \alpha\log\log x/\log x$), with no transition other than the slow growth of $\log H$.

The deficit arises by inserting Montgomery's singular-series average $\sum_{d\le h}\mathfrak{S}(d)(h-d) = \frac{h^2}{2}-\frac{h}{2}(\log h+\gamma+\log 2\pi-1)+o(h)$ into the second factorial moment at the microscopic window $h=\lambda\log x$. We state plainly, following external review, what this is and is not: the constant $\gamma+\log 2\pi-1$ is Montgomery–Soundararajan's, was identified in their framework long before this paper, and the statement is a *microscopic extrapolation and numerical test* of their second-moment formula at Gallagher's boundary—not a newly discovered second-order law. The hypothesis must also be quantitative: what is required is not the qualitative k-tuple conjectures but uniform Hardy–Littlewood estimates for all shifts up to $O(\log x)$, control of the average singular series at the $o(h)$ level, and bounds for prime powers, endpoint weights, and the variation of $\log t$ across the sampling range. Three further cautions are built into the statement: Gallagher's Poisson law is conditional, not a theorem about primes; the sampling measure for t is specified, since different weightings shift lower-order terms; and the error is claimed only as $o(1/\log x)$. The naive claim $\operatorname{Var}/\mathbb{E}=1$ fails at every accessible scale by 18–28%; the corrected law matches with no fitted parameter, and the observed residual $\approx +0.003$ is consistent with (but not evidence for) a genuine $1/\log^2 x$ term. (Numerical precedent: Sanchis-Lozano [26] fitted the Montgomery–Soundararajan variance shape at $x\sim 10^8$ in the mesoscopic regime $h\gg\log N$; the microscopic statement and test appear to be new, and we claim no more than that.)

	$x=10^9$		$x=4\cdot 10^9$	
λ	observed	predicted	observed	predicted
1/2	0.8241	0.8189	—	—
1	0.7910	0.7854	0.7998	0.7960
2	0.7527	0.7520	0.7670	0.7646
4	0.7201	0.7185	—	—

The residual $\approx +0.003$ is the anticipated $O(1/\log^2 x)$ term.

Following the fourth external review, the twin-race statement is split into its provable algebraic core and its conjectural clauses, stated through the logarithmic-mean functional $\mathcal{M}_x$ of the notation section.

Lemma 1 (Orientation elimination and class assignment)**.** *In $\psi_2(x)=\sum_{n\le x}\Lambda(n)\Lambda(n+2)$, the total contribution of terms in which n or $n+2$ is a proper prime power is, apart from $O(x^{1/3}\log^2 x)$ from cubes and higher powers, carried entirely by the patterns $(q^2-2,\, q^2)$ with q prime and q^2-2 prime: the mirror pattern $(q^2,\, q^2+2)$ is annihilated by $3\mid q^2+2$ for every prime $q>3$. Moreover, for every prime $q\ne 5$, $q^2-2\equiv 2$ or $4 \pmod 5$ (never 1), and for every odd q, $q^2-2\equiv 7 \pmod 8$. (The single exceptional term $q=5$, $(23,25)$, lands outside the twin-start classes altogether and is absorbed in the bounded error.)*

The proof is elementary (squares mod 3, 5, 8; the prime-power count). The lemma orients the race: prime-square contamination can enter only specific residue classes, and the model converts that into a drift prediction for the *unweighted* twin counts.

Conjecture 21 (Twin races modulo 5 and 8, and the prime-power contamination calculus; (b), apparently new)**.** *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod 5$. Let $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$ and*

$$T(x) \;=\; \frac{1}{2\log^2 x} \sum_{\substack{q \le \sqrt{x} \\ q^2-2 \text{ prime}}} \log q \, \log(q^2 - 2),$$

whose scale is governed by the constant C_7 of Conjecture 7. Then: (i) *[drift law] class* 1 *carries the systematic surplus T:*

$$\mathcal{M}_x\left(\frac{D_1 - T}{\sqrt{\pi_t}}\right) \longrightarrow 0 \qquad (x \to \infty),$$

i.e. the deterministic part of D_1 is exactly T, and the remainder is mean-zero noise at the random-walk scale under logarithmic averaging; (ii) *[zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term: $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4))/\sqrt{\pi_t}) \to 0$, and likewise for each of the three pairwise differences among classes $\{1, 3, 5\}$ mod 8 in clause (iv);* (iii) *[sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of $\{D_1 > 0\}$ exists and equals $\frac{1}{2}$, approached from above with a finite-x excess of order $\log\log x/\log x$: the pointwise drift-to-noise ratio is $T/\sqrt{\pi_t} \asymp 1/\log t$, and its* logarithmic average $\mathcal{M}_x(1/\log t) \sim \log\log x/\log x$ *carries an extra* $\log\log$ *(a scale the sixth external review corrected: an earlier draft wrote $1/\log x$). In contrast to Chebyshev's races, the twin race has no persistent leader;* (iv) *[mod-8 companion] since $q^2 - 2 \equiv 7 \pmod 8$ always (Lemma 1), the* entire *deterministic term falls on the single class $7 \pmod 8$: with $D_7(x) = \frac{1}{3}\sum_{a\in\{1,3,5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,*

$$\mathcal{M}_x\left(\frac{D_7 - 2T}{\sqrt{\pi_t}}\right) \longrightarrow 0$$

(twice the mod-5 term, undiluted), while classes $1, 3, 5$ are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) *[contamination calculus for balanced races] for a pattern $(n, n + d)$ and modulus m, call the race* balanced *if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the $\sqrt{x}$ scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the* contamination operator

$$\mathcal{C}_{d,m}(a; x) \;=\; \sum_{\textit{orientations } o\in\{(q^2-d,q^2),\,(q^2,q^2+d)\}} \; \sum_{\substack{q\le\sqrt{x},\ o\ \textit{prime-compatible} \\ \mathrm{start}(o)\equiv a\ (m)}} \Lambda(q^2)\,\Lambda(\textit{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which o survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the *transfer: the unweighted class counts are depressed by $\mathcal{C}_{d,m}(a;x)/\log^2 x$ relative to the clean classes, in the $\mathcal{M}_x$-sense of (i)—i.e. "exactly the removed mass," which requires the weighted identity, partial summation uniform in classes, higher-power bounds (exponents $r \geq 3$ contribute $O(x^{1/3+\varepsilon})$ against the operator's $\asymp \sqrt{x}$ and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and is the calculus's single analytic conjecture rather than a rule (the seventh review is right that this step, not the census, is the mathematical content). The operating scale should be kept in view throughout: the drift is $\asymp \sqrt{x}/\log^2 x$ against a count fluctuation $\asymp \sqrt{x}/\log x$, i.e. smaller by $1/\log x$ in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the cousin pattern $d = 4$ the calculus makes two predictions with no free choices: the orientation (q^2-4, q^2) is dead ($q^2 - 4 = (q-2)(q+2)$ is composite), the orientation (q^2, q^2+4) survives (no prime annihilates it), q^2+4 prime forces $q \equiv \pm 2 \pmod 5$, so the contaminated cousin-start class is $4 \pmod 5$ (of $\{2,3,4\}$) and—since $q^2 \equiv 1 \pmod 8$—the class $1 \pmod 8$ (of $\{1,3,5,7\}$): both deficits equal $T^{\mathrm{c}}(x) = \frac{1}{\log^2 x}\sum_{q\leq\sqrt{x},\ q^2+4 \text{ prime}} \log q \, \log(q^2+4)$ in the sense of the $\mathcal{M}_x$-clauses of (i)–(ii).*

Clause (v) is the answer to the fifth external review's structural question—what general principle do the mod-5 and mod-8 races instantiate—and the cousin predictions were derived first and tested second. At $x = 10^9$ the predicted drift is far inside the noise ($T^{\mathrm{c}}/\sqrt{\pi_{\mathrm{c}}} \approx 0.09$, as the $1/\log x$ law requires), so the sharp statistic is leadership under logarithmic averaging, calibrated against the corrected null of Conjecture 8(ii) (log-occupation $\frac{1}{2}$ with Gaussian spread $\sqrt{\log 2/\log x} \approx 0.18$ at 10^9, the occupation-fraction constant fixed by the seventh review): the class-4 deficit race led on the predicted side with log-density 0.99 (+2.7 null standard deviations), the class-1 mod-8 race with 0.92 (+2.3), both signs agreeing with the prediction, while the control differences among uncontaminated classes stayed within one noise unit of zero in log-mean. With the corrected constant this is moderately strong directional evidence—two independent projections each between two and three null standard deviations on the predicted side—though the races share underlying primes and the two statistics are positively correlated, so we still do not call it sharp confirmation; the family now has five projections of one mechanism (twin mod 5, twin mod 8, cousin mod 5, cousin mod 8, and the Goldbach lanes of Conjecture 25) plus two algebraic nulls (Conjecture 8 and the dead cousin orientation), which is the shape-of-agreement standard applied to the calculus itself.

An earlier draft of clause (iii) asserted a limiting sign density strictly between $\frac{1}{2}$ and 1; an external review correctly observed that this is inconsistent with the scale analysis—when drift/noise $\to 0$, a Gaussian model forces the sign density to $\frac{1}{2}$, and a nontrivial limiting density would require a persistent normalized bias that no mechanism here produces. A yet earlier draft wrote the error in (iv) as $O^*(\sqrt{\pi_t})$; the fourth review correctly objected that O-notation asserts a provable bound where only a conjectural fluctuation statement is meant, and the $\mathcal{M}_x$-clauses above say exactly what is conjectured and no more. Two model assumptions remain open and are stated as such: the passage from the Λ-weighted count to unweighted twin counts is a partial-summation argument whose error terms need writing out (higher prime powers are disposed of by Lemma 1); and the *zero-oscillation caveat*—the drift law treats the oscillatory explicit-formula contributions to class differences as having vanishing logarithmic mean, the exact analogue of the Rubinstein–Sarnak GRH-plus-linear-independence framework [6], and a conspiracy among hypothetical zeros could in principle mimic or cancel part of the drift; clause

(i) is conjectured on the standard side of that hypothesis.

The mechanism is the twin analogue of the classical explanation of Chebyshev's bias: prime-square terms contaminate the classes that can contain them. (Biases of twin primes in residue classes have been studied before—Sahoo [27] for the mod-4 race, and [28] for prime pairs versus isolated primes—but we could find no prior statement of the mod-5 race, the (q^2-2, q^2) mechanism, or the quantified surplus below.) Mod 3 kills the pattern (q^2, q^2+2) outright, an amusing local accident that halves the mechanism and (as a mod-3 computation showed, overturning a first guess of a quadratic-residue bias against class 4) concentrates the surplus entirely on class 1. Data to $4 \cdot 10^9$: class counts (3980017, 3982505, 3981922), D_1 within one noise unit of T throughout, no persistent leader, and the control race $\pi_{t,2} - \pi_{t,4}$ wandering like a fair coin. Mod-8 data to 10^9: class counts (856684, 855807, 856046, 855967) for $a = 1, 3, 5, 7$; $D_7 = +212$ at 10^9 against predicted $T = +254$ with noise $\sim$1068, positive through most of the range (log-density 0.775), and the three pairwise $\{1, 3, 5\}$ controls all within one noise unit of zero—the two projections of the mechanism behave coherently, none of it provable at present.

Conjecture 22 (Least primes in progressions; (a) for part (i) [20, 21, 30, 29], (b) for part (ii))**.** *Let $p(a, q)$ be the least prime $\equiv a \pmod q$ and $U(a, q) = \mathrm{Li}(p(a, q))/\varphi(q)$. Then:* (i) *$U \Rightarrow \mathrm{Exp}(1)$ marginally as $q \to \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \le n} 1/k$), the order-statistics form of Wagstaff's $\max_a p(a, q) \sim \varphi(q) \log^2 q$;* (ii) *[canonical deficit, dyadic-average form] define, with no reference to any control model, $\Theta(q) = \big(1 - \mathbb{E}_a[U(a, q)]\big) \log q$. The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime q may be too strong, since arithmetic fluctuations in q could persist (the sixth review's caution):*

$$\frac{1}{\#\{q \in [Q, 2Q] \ \textit{prime}\}} \sum_{\substack{Q \le q \le 2Q \\ q \text{ prime}}} \Theta(q) \ \longrightarrow \ \Theta \ > \ 0 \qquad (Q \to \infty).$$

(The formal claim is deliberately model-free: existence and strict positivity of the dyadic-average limit. The stronger reading—that Θ exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution *of $\Theta(q)$ over the dyadic block is left open: an earlier draft asserted $\mathrm{sd}(\Theta(q)) \to 0$, but fluctuations driven by the arithmetic of q and of $q-1$ (through the singular-series pair terms $\mathfrak{S}(kq)$ and the class structure mod q) may well persist at bounded size, so $\Theta(q)$ need not concentrate around its dyadic mean, and we do not conjecture that it does.*

The decomposition $\Theta(q) = \theta_{\mathrm{disc}}(q) + \theta_{\mathrm{corr}}(q)$ is a *diagnostic*, not a canonical invariant—the fifth external review is right that its two parts move if the benchmark moves, and clause (ii) is therefore stated through the model-free $\Theta(q)$ alone. The named benchmark (the parity-aware Cramér–Bernoulli control: candidates a, $a+q$, $a+2q, \ldots$ independently "prime" with hazard $h_i = (q/\varphi(q)) \cdot s_i/\log(a+iq)$, $s_i = 2$ for odd candidates and 0 for even, matching density and parity) defines $\theta_{\mathrm{disc}}(q) = (1 - \mathbb{E}_a[U^{\mathrm{model}}]) \log q$ exactly for each q, and $\theta_{\mathrm{corr}} := \Theta - \theta_{\mathrm{disc}}$ is the part of the measured deficit that the control cannot produce; it is this diagnostic that localizes the anomaly in the *ordering* of the primes (Remark 1). Measured on prime moduli $q \in [1500, 6000]$: $\Theta = 1.671 \pm 0.009$, of which the control accounts for 0.847.

One further deterministic baseline must be subtracted before any of the residue is called an anomaly, and the eighth external review is right to demand it: for prime q, the primes

$p < q$ occupy *distinct* classes mod q—an injective initial phase that no exchangeable allocation reproduces. Its size has a closed form in index time (where $\mathrm{Li}(p_i) \approx i$, so U is the first-hit index over φ): forcing the first $\pi(q)$ arrivals to be collision-free and leaving the rest exchangeable gives $\mathbb{E}[U] = 1 - \pi(q)^2/2\varphi^2 + O(\pi(q)/\varphi^2)$, hence

$$\Theta_{\mathrm{inj}}(q) \;=\; \frac{\pi(q)^2}{2\varphi(q)^2}\,\log q\,(1+o(1)) \;=\; \frac{1+o(1)}{2\log q},$$

closed form $0.092,\ 0.082,\ 0.074$ at $q = 1499,\ 3001,\ 5987$, with Monte Carlo agreeing within sampling error and the exchangeable control at 1.000. The injective phase is therefore real but an order of magnitude too small to be the explanation: it accounts for $\approx$0.08 of the measured $\theta_{\mathrm{corr}} = 0.824 \pm 0.009$, and carries a $1/\log q$ decay that the flat measured values do not show. (The review suggested the effect sits at the full deficit scale because the *affected-class fraction* is $\pi(q)/\varphi \asymp 1/\log q$; but the deficit is the affected fraction times the collision probability an exchangeable model would have suffered in that phase, $\asymp \pi(q)/2\varphi$, giving $1/2\log^2 q$ in $\mathbb{E}[U]$, not $1/\log q$.) The injective phase is in any case the $y < q$ head of the pair-correlation expansion already registered below—collisions require $q \mid p_2 - p_1$, impossible before $y = q$—so subtracting it is the first term of the stated programme, not a rival explanation. The residue $\theta_{\mathrm{corr}} - \Theta_{\mathrm{inj}} \approx 0.74$ remains the unexplained ordering anomaly.

The sixth external review supplies the derivation route this conjecture has lacked, and we adopt it as the slot's stated programme. Let $V_q(y)$ be the number of reduced classes mod q not yet visited by a prime $\le y$; then $\mathbb{E}_a[U] = \int_0^\infty \overline{V_q(y_\tau)}/\varphi(q)\,d\tau$ with the Li-time change y_τ defined by $\mathrm{Li}(y_\tau) = \tau\varphi(q)$, and inclusion–exclusion expands $V_q(y)$ in prime-tuple correlations *within* a class:

$$\mathbb{E}\binom{N_a}{2} \text{ summed over } a \;=\; \#\{p_1 < p_2 \le y :\ q \mid p_2 - p_1\} \;\approx\; \mathrm{Li}_2(y) \sum_{k\ge 1} \mathfrak{S}(kq),$$

so the first correction to the exchangeable occupancy model is carried by the Hardy–Littlewood singular series of gaps divisible by q. Evaluating this expansion to second order, and comparing it with the measured Θ, is the concrete open computation that would either derive the ordering anomaly from pair correlations or prove it lies deeper; either outcome retires Question 1's mystery one level.

The stratified experiment behind these numbers—demanded in substance by the fourth external review—separates moduli by factorization type in the matched range $q \in [1500, 6000]$: over 40 *prime* moduli, $\theta_{\mathrm{disc}} = 0.847$ (essentially constant across q) and $\theta_{\mathrm{corr}} = 0.824 \pm 0.009$; over 40 7-*smooth* moduli in the same range, $\theta_{\mathrm{disc}} = 4.32 \pm 0.22$ (the large $q/\varphi(q)$ hazards make discreteness dominant) while $\theta_{\mathrm{corr}} = -2.32 \pm 0.22$ reverses sign. The comparison with Leung [29], raised by the third review, is thereby given its answer-shaped experiment: Leung derives the exponential law under a uniform Hardy–Littlewood hypothesis and reports discrepancies for *smooth* moduli, and our smooth stratum indeed behaves anomalously—but the positive ordering term survives, with small error bars, exactly where smooth-modulus effects cannot reach. Whether θ_{corr} tends to a constant, and the exact functional form in $1/\log q$, remain open: data at $q \le 6000$ cannot distinguish $1/\log q$ from $(\log\log q)/\log q$-type refinements—the aggregate $\theta \approx 1.5$–1.9 over $q \in [10^2, 6 \times 10^3]$, with $\mathbb{E}_a[U]$ bottoming near 0.737 around $q \approx 200$ and recovering through 0.776 by $q \approx 6000$, is calibration, not a determination of the form. The averaging family is now fixed as: all moduli of the stated type in dyadic ranges of q, per stratum.

Remark 1. *The deficit is reproducible and is* not *an artifact of the time-change: a Cramér pseudo-prime control (independent Bernoulli "primes" with the correct densities and parity) shows roughly half of it—the discreteness of large per-candidate hazards—while the other half vanishes under any reshuffling of the actual prime residues (iid labels, or a random permutation of the true residue sequence, both restore* $\mathbb{E}[U] = 1.00$*). The ordered sequence of primes fills residue classes measurably faster than any exchangeable model, and the stratified experiment above shows the effect is not a smooth-modulus artifact: it is largest and cleanest on prime moduli. We measure* θ_{corr}*; we cannot yet derive it. It is the deepest unexplained number produced by this exercise.*

Conjecture 23 (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic is (a) [22], the vertical multibase law apparently new)**.** *With* $q_p(a) = (a^{p-1} - 1)/p \bmod p$ *and* $q_p = q_p(2)$*, five clauses:* (0) *[structure, classical]* $q_p(ab) \equiv q_p(a) + q_p(b) \pmod p$ *(the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively* dependent *bases the vector* $(q_p(a_1), \ldots, q_p(a_r))/p$ *is confined to the rational subtorus cut out by the relations;* (iv) *[multibase equidistribution, the claimed law] for multiplicatively* independent *bases the vector equidistributes on the full torus as p varies—the* vertical *joint law, complementing the fixed-p, varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i);* (v) *[simultaneous Wieferich] the expected count of* $p \le x$ *with* $q_p(2) = q_p(3) = 0$ *has convergent sum* $\sum 1/p^2$*: at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad's), and the empirical list is empty to* 10^7*. Three further clauses on the base-2 quotient alone:* (i) *[global equidistribution]* q_p/p *equidistributes on* $[0, 1)$ *with discrepancy at exactly the calibrated random-model scale: in its sharp form,*

$$\limsup_{x\to\infty} \frac{\sqrt{\pi(x)}\, D_{\mathrm{KS}}(x)}{\sqrt{\log\log \pi(x)}} \;=\; \frac{1}{\sqrt{2}},$$

the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (an earlier draft asked $O(\pi(x)^{-1/2})$, stronger than the random model supports*, the same over-demand that killed the Mertens conjecture, caught by external review) and, per the fifth review, no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise;* (ii) *[shrinking targets] for every fixed* $K \ge 1$*, and uniformly for* $K = K(x) \le \log x$*,* $\#\{p \le x : \; q_p < K\} \sim \sum_{p \le x} \min(1, K/p)$*—the fixed-K case down to* $K = 1$ *is what governs the zero event and does NOT follow from (i), since* $\{q_p = 0\}$ *is a target of shrinking measure* $1/p$*;* (iii) *[Wieferich count, fluctuation-calibrated, with the probability space declared]* $W(x) = \#\{\text{Wieferich } p \le x\}$ *is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms (the seventh review is right that an unqualified CLT for a fixed sequence has no meaning): the* deterministic envelope conjecture

$$\limsup_{x\to\infty} \frac{W(x) - \log\log x}{\sqrt{2 \log\log x \, \log\log\log\log x}} \;=\; 1$$

(the iterated-logarithm scale at variance $V(x) \sim \log\log x$*; an earlier draft's* $O_\varepsilon(V^{1/2+\varepsilon})$ *was weaker than the model's exact prediction), and the* randomized central limit law, with a sampling window wide enough to matter*: sample u uniformly from* $[1, \log\log X]$ *and set* $x = \exp(\exp(u))$ *(equivalently, weight x by* $d(\log\log x)$ *on* $[\mathrm{e}^{\mathrm{e}}, X]$*—the almost-sure-CLT weighting); then* $(W(x) -$

$u)/\sqrt{u}$ converges in distribution to $N(0,1)$ as $X \to \infty$. An earlier draft sampled x log-uniformly from a block $[X, X^A]$; the eighth external review correctly observed that this is vacuous as a CLT: across such a block the expected number of new *Wieferich events is $\log\log X^A - \log\log X = \log A + o(1) = O(1)$, so the block-sampled statistic is frozen at the historical value $(W(X) - \log\log X)/\sqrt{\log\log X}$ and cannot converge to a fresh normal. The repaired form samples a window on which the variance itself,* $\log\log x$*, sweeps an interval of diverging length—the standard almost-sure-CLT resolution of exactly this degeneracy. Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)'s discussion records.*

Two corrections from external review are absorbed here. First, an earlier draft derived (iii) from (i) with "hence"; that inference is invalid—a square-root-size error in the aggregate distribution is astronomically larger than $\log\log x$, and only the shrinking-target clause (ii) at bounded K speaks to the zero event. Second, the earlier fluctuation claim $O(1)$ violated this paper's own calibration principle (criterion of demanding exactly the fluctuations the model earns): with event probabilities $1/p$ the count has standard deviation $\sim \sqrt{\log\log x}$, and the statement now says so. We also retract any location prediction for a third Wieferich prime: the expected count reaching 3 "at doubly exponential heights" is an expected-count statement whose additive constant is uncalibrated, not a forecast. Consistency data: $\{1093, 3511\}$ below 10^8 against expected ≈ 2.7; $\sqrt{n}\,\mathrm{KS} = 0.82,\ 0.82,\ 0.52,\ 0.63$ at $x = 10^5, \ldots, 10^8$ (bounded, no drift); census at $K = 100$: observed 169 against model 161.2.

The multibase clauses were verified at 10^7 (664,577 primes): the homomorphism (0) held *exactly* on every spot-checked prime; for the independent pair $(2,3)$, correlation $+0.0023 \pm 0.0012$, a 20×20 joint occupancy χ^2 of 441 on 399 degrees of freedom ($+1.5\sigma$), joint small-quotient census 46 against model 41.2 ($z = +0.75$), and no simultaneous Wieferich prime. Two positions taken deliberately: the vertical multibase law (iv) sides with the Crandall–Dilcher–Pomerance accounting against Gras's competing probabilistic model (which predicts $q_p(a) = 0$ *rarer* than $1/p$, hence possibly finitely many Wieferich primes even for a single base)—the disagreement is precisely what gives clauses (iii)–(iv) content; and the horizontal literature (fixed p, base varying: Ostafe–Shparlinski's joint distributions and the Fermat-quotient-matrix statistics of Cobeli–Zaharescu and coauthors) is the attributed neighbour whose vertical counterpart is what this slot claims.

Conjecture 24 (Conjecture F as a family; (a), [1], cf. [23])**.** *For odd A let $Q_A(N) = \#\{n \le N :\ n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then, for every fixed $B > 0$, $Q_A(N) = C(A)\, I_A(N)\,(1 + o(1))$ uniformly over odd $1 \le A \le (\log N)^B$, with the residual field obeying the analogue of Conjecture 16(ii), kernel derived and probability space specified: for t uniform on $[N, 2N]$ and window length $H \le N^{o(1)}$, let $Q_A(t; H) = \#\{t < n \le t + H :\ n^2 + n + A \text{ prime}\}$ (mean $\asymp H/2\log N$, which sets the regime). Two members share the index window, so with $C(A, A') = C(x^2 + x + A,\ x^2 + x + A')$,*

$$\mathrm{Cov}_t(Q_A, Q_{A'}) \ \sim\ [C(A, A') - C(A)\, C(A')]\, \frac{H}{4\log^2 N},$$

with $C(A, A') = 0$ (hence negative *correlation) for the locally exclusive pairs. This is the* same-index *contribution only (the seventh review's correction of an earlier overclaim): the full covariance adds the off-index pinned sum $\sum_{|h| \le H}(H - |h|)\,[C(f_A(x), f_{A'}(x+h)) - C(A)C(A')]/4\log^2 N \cdot \log^{-2} N$, the two-parameter analogue of Conjecture 12's $G(H)$, whose evaluation is this family's open component. For $H/\log N \to \lambda$ the counts are jointly Poisson with local exclusions, and for*

$H/\log N \to \infty$ each fixed finite collection of studentized counts is asymptotically jointly normal, all correlations of size $\asymp 1/\log N$ making both kernels finite-N corrections to independence, exactly as at Conjecture 16(ii). (Uniformity over a range growing with N is the substantive claim; as an external review noted, uniformity over any fixed *finite set of A is logically automatic from the individual asymptotics.) Among odd $A \le 199$, Euler's $A = 41$ has the largest constant, $C(41) = 6.6395463\ldots$ (Cohen's high-precision value [31]).*

Numerical evaluation of the kernel (all odd $A, A' \le 99$ at $N = 10^6$: 1,225 pairs, of which 289 locally exclusive) returned a structurally clean answer: the positive correlations of admissible pairs and the negative correlations of exclusive pairs *cancel*, mean off-diagonal $\rho = -0.003$, so the derived cross-member kernel is null and the members are asymptotically independent after centering. The observed profile variance $\approx$0.37 over the same range is therefore *not* a common-mode effect; it must be diagonal—each $\mathrm{Var}(Q_A)$ sub-Poisson through within-sequence pair correlations, the quadratic-family analogue of the Montgomery–Soundararajan deficit (Conjecture 20)—and part of the spread is truncation noise in the constants themselves (the $\pm 10^{-3}$ wobble moves each z_A by $\sim$0.3). Deriving the diagonal law and the off-index pinned sums is the family's remaining open component—and, as two reviews have stressed, the more important one; the numerical cancellation of the *same-index* cross-member kernel is a finite computation over $A \le 99$, which localizes the deficit off the same-index diagonal without proving that the averaged kernel or the off-index contributions vanish—an analytic classification of the sign of $C(A, A') - C(A)C(A')$, and the evaluation of the two-parameter pinned sums, remain open. At $N = 10^6$: mean ratio 1.0006, standard deviation 0.0027, $\max |z| = 1.92$, rank correlation 0.99988; for $A = 41$, observed 261,080 against predicted 261,017.6 ($z = +0.12$, using Cohen's constant). A cautionary note from the referee audit belongs here: our own truncated Euler product for $C(41)$ at cutoff $2 \cdot 10^5$ reads 6.64092—wrong in the fourth significant decimal—because these conditionally convergent products drift at the 10^{-3} scale over practical cutoffs. Quoted digits for all quadratic-family constants in this paper are limited by the stated truncation wobble, not by the last printed digit. The entire 10^8-bit primality dataset compresses to one product formula per A—the description-length virtue in its purest form.

Conjecture 25 (The Goldbach lane race; apparently new)**.** *For $n \equiv 2 \pmod 4$ let $R_1(n)$, $R_3(n)$ be the ordered Goldbach representation counts in the lanes $p \equiv q \equiv 1$ and $p \equiv q \equiv 3 \pmod 4$ (Conjecture 15), and $D(n) = R_3(n) - R_1(n)$. Prime-square terms $n = q^2 + p$ (q odd prime) land* exclusively *in the $(1,1)$ lane, since $q^2 \equiv 1 \pmod 4$ and then $n - q^2 \equiv 1 \pmod 4$ automatically. Consequently the $(3,3)$ lane leads on average: with*

$$D_{\mathrm{sys}}(n) \;=\; \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime},\ q \le \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \, \log(n - q^2), \qquad \bar{\ell}(n) \;=\; \frac{n-6}{\displaystyle\int_3^{n-3} \frac{dt}{\log t \, \log(n-t)}},$$

(the analytic mean of $\log p \log(n-p)$ under the lane profile; the empirical lane mean is its estimator). Write $\mathbb{E}_x$ for the average over n drawn from $\{n \equiv 2\ (4),\ n \le x\}$ with weight proportional to $1/n$ (the logarithmic sampling measure, now explicit). Then: (i) *[drift law] $\mathbb{E}_x[D - D_{\mathrm{sys}}] = o(\mathbb{E}_x[D_{\mathrm{sys}}])$ as $x \to \infty$: the $(3,3)$ lane leads on average by exactly the square contamination;* (ii) *[weighted mean and internal null lane] $\mathbb{E}_x[D_{\mathrm{sys}}]$ is given by the Hardy–Littlewood (Conjecture H) local model for $n - q^2$, whose local factors vary with n as the fourth external review stressed—most drastically at $p = 3$: for $n \equiv 1 \pmod 3$, $3 \mid n - q^2$ for* every *prime $q > 3$, so the contamination collapses to exactly two boundary families—the $q = 3$ term*

(when $n-9$ is prime) and, as the sixth external review noted against an earlier "single term" phrasing, the sparse exceptional family $n-q^2=3$ (when $n-3$ is itself a prime square, so that the divisible-by-3 value is *3)—both negligible in logarithmic mean, so the race carries essentially no drift on the subprogression* $n \equiv 10 \pmod{12}$, *while the drift concentrates on* $n \not\equiv 1 \pmod 3$*: an internal null lane, predicted by the same mechanism that predicts the lead;* (iii) *[sign density] the per-n drift-to-noise ratio $\kappa(n)=D_{\mathrm{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$ (where $J(n)=\int_3^{n-3} dt/(\log t\log(n-t))$, so that $\mathfrak{S}(n)J(n)$ is the total ordered representation count) satisfies $\kappa(n)\asymp c(n)/\log n\to 0$; under the Gaussian fluctuation model the logarithmic density of $\{D>0\}$ is therefore $\frac{1}{2}$ in the limit, with the finite-x sign fraction predicted to be $\mathbb{E}_x[\Phi(\kappa)]$ (Φ the standard normal distribution function)—decaying, but far from $\frac{1}{2}$ at any accessible height.*

Derivation status: the identity behind D_{sys} is the Λ-weighted lane symmetry minus its prime-power part; the transfer to unweighted counts is a partial-summation argument whose error terms (higher prime powers, which enter only at the $n^{1/3}$ scale, and the variation of the weight across the lane) remain to be written out—the same open transfer flagged at Conjecture 21—and clause (iii) inherits the zero-oscillation caveat stated there.

This is the Goldbach-partition analogue of Chebyshev's bias, produced by the same explicit-formula mechanism as Conjecture 21 but through q^2+p patterns rather than twin patterns; the single-sign prediction (every square contamination falls into one lane) makes it cleaner than the classical race. We could find no prior statement. It replaces, at the suggestion of an external review, a restatement of the Lemke Oliver–Soundararajan mod-3 law that formerly occupied this slot and carried no new content (their conjecture remains cited at Conjecture 16's family level and its verification data—$\hat{c}$ drifting $0.400\to 0.370$ over $10^7\to 4\cdot 10^9$ with $(1,1)/(2,2)$ symmetry 0.99989—remain in the repository).

Verification. Over 500 log-spaced samples $n\le 10^8$: mean $D=76.5$ against mean empirical $D_{\mathrm{sys}}=64.5$, the $(3,3)$-lane lead significant at 5.2 standard errors of the mean. Clause (ii)'s weighted model, computed in presieve-exact form (trial division of each $n-q^2$ to 10^3 with the exact Mertens normalization), reproduces the empirical contamination with no fitted parameter: model mean 66.3 against empirical 67.8 (ratio 0.98) on a 400-sample run. The internal null lane behaves as predicted: on $n\equiv 1$ (3) (123 samples) the model and empirical D_{sys} both vanish and the measured lead drops to 31 ± 23 (consistent with zero), while on $n\not\equiv 1$ (3) (277 samples) the lead is 88 ± 21 against predicted contamination 96—the drift lives exactly where the mechanism puts it. Clause (iii): computed $\mathbb{E}_x[\Phi(\kappa)]=0.572$ (κ ranging 0–0.64 with mean 0.18) against observed sign fraction 0.588; weak, and quantitatively the weakness the model requires.

7 The adversarial audit

Each statement above was attacked three ways.

Layer 1: literature. Five independent literature searches combed OEIS, arXiv, and the standard references for prior art, producing the novelty labels and the attribution section below. This layer also falsified one historical claim in our working notes (the $k\ge 1$ Stern list of Conjecture 14 was described as partly new; it is OEIS A060003 verbatim, and 1493 is the largest known Stern prime); the notes were corrected on the record. The layer also produced the audit programme's own worst failure: a snippet-depth search cleared the primorial-twin statement that Lillie [5] had in fact already made (Finding 2). All novelty searches for the statements admitted in the present revision (the Conjecture 1 and 5 family lifts, the chain of Conjecture 6, the null race of Conjecture 8, the factorial twins of Conjecture 10) were therefore run at abstract depth: the nearest-neighbour papers were read, not skimmed from search results.

Layer 2: scripted refutation. Every falsifiable-by-instance statement was pushed at least a decade past its original bound: exception hunts across $(10^8, 10^9]$ for Conjectures 13, 14, 15, 17; the uniformity of Conjecture 16 stressed on $d \leq 6000$; the statistical laws re-tested at $4 \cdot 10^9$ (Conjectures 18–21, 25, and the quintuplet count of Conjecture 4); the recovery clause of Conjecture 22 tested on fresh moduli $q \in (3000, 6000]$. Outcome: one kill (Finding 3); no other statement moved. The statements admitted or upgraded in the present revision were verified at production scale before admission: the Conjecture 1 profile across 150 shifts; the chain of Conjecture 6 over five decades to 10^7; the null race of Conjecture 8 at 10^7; factorial twins to $n = 700$; the CRT-exact factorization of Conjecture 11 through a joint period of 1.2×10^8 with the scan extended to $n = 6000$; the covariance kernels of Conjectures 16(ii) and 24(ii) evaluated over 870 and 1,225 pairs; the stratified experiment of Conjecture 22 on 80 moduli; and the weighted drift and internal null lane of Conjecture 25 on 400 fresh samples. A fifth review round added three more derive-first-test-second runs: the moving-window randomization of Conjecture 16(ii) (2000 windows: empirical correlation matrix matches the predicted kernel entrywise at 0.86); the cousin-race predictions of the contamination calculus, Conjecture 21(v), at 10^9 (leadership log-densities 0.99 and 0.92 on the predicted sides); and the singular-series waiting-time clause of Conjecture 19(iii) (regression coefficient -0.466 against the predicted $-\frac{1}{2}$). A sixth round corrected four formulation errors caught by review—the occupation clock of Conjecture 8(ii) (arcsine only in the natural clock; ergodic $\frac{1}{2}$ under logarithmic averaging), the $\log\log x/\log x$ sign-excess scale of Conjecture 21(iii), the $n \geq 4$ restriction and $n - q^2 = 3$ exceptional family in Conjectures 10(i) and 25(ii), and the Gaussian-regime conditions $H/\log^2 x \to \infty$ (resp. $H/\log N \to \infty$) in Conjectures 16(ii) and 24(ii)—and added two derive-and-test items: the exact rational CRT identity of Conjecture 11(ii), and the moment law for the constants $C(d)$ of Conjecture 1(iii) (derived mean 2.7447 and sd 1.6835 against measured 2.7434 and 1.6726). A seventh round—the wholesale replacement of every remaining standalone benchmark—added ten new statements, each searched at neighbourhood depth by two independent audit agents before admission and verified at production scale: the cubic family moments and uniformity (294 constants, 57 count profiles); the triplet and sexy-pair races at 10^9 and the Stern lanes on 2,500 samples at 10^8; the Fibonacci–Lucas joint scan to $p \leq 10^4$, where the search *refuted* the draft completeness clause via OEIS A080327's index 148091 before publication—the audit working exactly as designed; the pinned average $G(H)$ computed exactly to $H = 3000$; the boundary trichotomy with two new lanes counted at 10^6; the least-summand law on 2,000 samples; the orientation-resolved twin-member profile on 150 samples; the race autocorrelation and running-maximum measurements at 10^9 against simulated nulls; and the multibase quotient tests on 664,577 primes.

Layer 3: clean-context replication. Independent replications, working from the bare statements alone without access to our code or notes, recomputed constants from definitions and recounted sequences with their own implementations. All recomputed constants landed inside the stated truncation error; all recounts agreed (one replication initially disagreed at $z \approx -22$, traced by the replicator itself to a bug in *its* sieve—every prime $n \leq Q$ had been struck out by its own modulus—after which it agreed at $z < 1.6$). The same layer reproduced the Fibonacci deficit of Remark after Conjecture 9 and the ordering anomaly of Remark 1, confirming both are properties of the primes rather than of our code.

We draw one methodological conclusion. Every fatal mathematical error the audit programme has caught in its own output was not statistical but *algebraic*: factorizations or congruence collapses living on density-zero or positive-density-but-structured subsequences (Findings 3 and 1, the $k = 4$ parity branch of Conjecture 5, the inadmissible naive chain of Conjecture 6); and the one fatal *bibliographic* error was a search read at snippet depth (Finding 2). Probabilistic sanity checks, however extensive, integrate over density and cannot see such families; only structural attack surfaces (here: "test the claimed exception census on special subsequences") find them. An audit that only re-runs the original verification at a larger bound would have declared the false version of Conjecture 13 sound.

8 Attribution and novelty

The calibration layer, previously stated in essence and retained inside the slots as attributed remarks, with our contribution being the uniform quantification, computed constants, and re-verification: C$1_{d=2}$ (OEIS A080149; [32]); $a = 2$ inside C2; the Chernick chain inside C3 [10]; C6's length-2 sub-chain (constant catalogued as OEIS A188596); $\{p, p^2 - 2\}$ inside C7 (OEIS A062326); the single-sided Fibonacci/Lucas laws inside C9 [7] and the twin object OEIS A080327; C10(ii) (OEIS A088054, the intersection comment); the factorial-prime law inside C12 [4]; C13(iii) [1] and Cunningham's cuban observation (OEIS A002407); the Stern list inside C14 (OEIS A060003); the $(3, 3)$-lane conjecture inside C15 [13] and the extremal theory of least summands [12]; C16-core (uniform Hardy–Littlewood B, cf. [14]); the twin-member basis conjecture and integral kernel inside C17 [9]; the record framework inside C18 [15]; the single-base Fermat-quotient heuristic inside C23 [22], with the Eisenstein–Lerch homomorphism classical; C24-core [1, 23]. Retired former occupants with their own history: the primorial twins of slot 10, Lillie [5] (Finding 2); the primorial benchmark before that and the Lemke Oliver–Soundararajan restatement of slot 25, [4] and [24]; the Sophie Germain statement of slot 6 and the quadratic triple of slot 8. Framework attributions for claimed clauses that are instances inside another theory: Kowalski's Euler-product limit laws [11] for C2(ii); Montgomery–Soundararajan [19] for the reductions behind C12 and C20; the Korevaar–te Riele mean-value-one theorem (arXiv:0806.1667) as the linear antecedent of C1(iii) and C2(i).

For the claimed layer we retain the review's grading discipline. *Bibliographic novelty only*: the constant of C5(iii). *Conceptual novelty*—a mechanism, law, family uniformity, or dichotomy we could not find in the literature: C1's uniform quadratic de Polignac family with its derived constant-moment law; C2's cubic constant-distribution law (instance-level inside Kowalski's framework, so graded); C3, C7 and C14: fresh contamination-calculus predictions (triplet race, two-orientation matrix, and the norm-form Stern lanes with provable null classes); C4's power-obstruction ladder and C13's boundary trichotomy (Cunningham's cuban case attributed); C5's complete twin-base family analysis; C6's alternating cyclotomic chain; C8's null-mechanism contrast prediction; C9's rank-disjointness analysis and quantified naive-constant refutation; C10's window-rigidity and joint-fluctuation clauses; C11's entanglement-aware local product with its exact CRT identity; C12's pinned singular-series reduction of the pair variance; C15's time-changed least-summand law and deficit Θ_G; C16(ii)'s derived covariance kernel (and its C24(ii) family analogue with the computed null cross-kernel); C17's orientation-resolved twin-member singular series; C18's measured race sub-diffusivity; C19's liminf constant and $\mathfrak{S}^*$-waiting-time clause (against the slope-1 laws of Shanks, Wolf, and Kourbatov–Wolf [25]); C20's microscopic variance law (a sharpening of [19], with the mesoscopic numerics of [26] as precedent); C21's race mechanism and calculus (against the twin-bias literature [27, 28]); C22(ii)'s canonical deficit with its stratified differentiating experiment; C23's vertical multibase subtorus law; and C25's Goldbach lane race. Among these we consider the contamination calculus with its six verified projections and three nulls (C3, C7, C8, C14, C21, C25), the two ordering deficits (C15, C22(ii)), the pinned-average reduction C12, and the sub-diffusivity measurement C18 the paper's most genuinely new content, and Question 1 its best export.

Of the structural programmes raised by the fifth external review, three have since been folded into the statements themselves (the boundary-factorization principle, now Conjecture 13; the quantitative twin-member law, now Conjecture 17; multibase Fermat quotients, now Conjecture 23(iv)–(v)), and two remain recorded as open directions: the admissibility dynamics of cyclotomic composition words (Conjecture 6), and centering-and-scaling extreme-value laws for record gaps, where Kourbatov's Gumbel-type framework [15] already occupies part of the ground and any contribution of ours would need separating from it. The asymptotic form of the pinned average $G(H)$ (Conjecture 12) and the derivation of the race repulsion constants ρ_k from the Lemke Oliver–Soundararajan series (Conjecture 18) join them as the replacement round's own open exports.

Question 1. Determine $\theta(q)$ of Conjecture 22: derive, from the Hardy–Littlewood correlations or otherwise, the first-order deficit of $\mathbb{E}_a\ \mathrm{Li}(p(a,q))/\varphi(q)$ below its random-model value 1—equivalently, explain quantitatively why the primes in their natural order occupy residue classes faster than any exchangeable resampling of themselves.

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